I. Earth – origin and composition

Fill in the blanks

1. The term soil is derived from Latin word ________________ which means ___________.
2. Pedology refers to ____.
3. Edaphalogy refers to ____.
4. _______________________ is considered as the father of Soil Science,
5. Soil is composed of three components *viz.*,______________, ______________ and ___________.
6. The solar system consists of the sun, _______ planets, and their ____ satellites, and a belt of asteroids.
7. The asteroids are otherwise called as __________________.
8. ______, _______, _______and ____ are the inner planets which are otherwise called as ________________.
9. The outer planets are otherwise called as ________________.
10. The galaxy in which we live is ________________________.
11. The solid zone of earth's sphere is termed as ____________________.
12. The hydrosphere refers to _______________________________________.
13. The thickness of Crust, Mandle and Core is __________, ___________& _________ respectively.
14. Most of the hard, naturally formed substance of the earth is referred to as ___________.
15. Out of the elements known, ____ elements are sufficiently abundant as to constitute 98.6 % of the earth's crust.
16. ________ and ________ are the non-metallic elements that are in greatest abundance and comprise nearly three-fourth of the total composition of the crust.
17. _________________ rocks occupy about 95 % of the earth's crust as a whole.
18. ____________ rocks occur to the extent of 74 % at or near the surface of the earth (upper 5 km).
19. It is evident that, the density ___________ from the crust to the core.
20. The density of the earth as a whole is _______, whereas the density of the common rocks varies from __________ to ___________.

Differentiate the following

1. Lithosphere Vs Hydrosphere
2. Core Vs Crust
3. Terrestrial planets VS Great planets
4. Inner Core Vs Outer core
5. Atmosphere Vs Pedosphere

Write a short note on the following

1. Composition of earth's crust
2. Interior of the earth
3. Origin of earth
4. Major divisions of earth's sphere
5. Relative abundance of rocks in earth's crust

Answers - Earth – origin and composition

Fill in the blanks

1. Solum, floor
2. Study of soil from the point of view of origin, classification etc.
3. Study of soil from the point of view of growing crops.
4. Dr.V.V.Dokuchaev
5. Solid, liquid and gaseous
6. 9, 31
7. Planetoids
8. Mercury, Venus, Earth and Mars, terrestrial planets
9. Great planets
10. Milky Way
11. Lithosphere
12. Sphere of water surrounding the earth
13. 5 – 56 km, 2900 km, 3500 km
14. rock
15. 8
16. oxygen and silicon
17. Igneous
18. Sedimentary
19. increases
20. 5.5 Mg / m^3 , 2.6 Mg / m^3 to 2.7 Mg / m^3

II. Soil Forming Rocks

Fill in the blanks

1. _________ is the very light weight rock which has a lower specific gravity and floats on water.
2. Igneous / Sedimentary / Metamorphic rocks are otherwise called as ______________________.
3. ___________ is an example of acid / basic igneous rock.
4. Magma refers to ____.
5. Lava refers to ____.
6. ________________ are the few examples of mono-mineralic rocks.
7. ____________________ is an example of coarse-grained / fine-grained rocks.
8. __________________ rocks are characterized by non-laminar massive structure and, on the whole, make up 95 % of the earth's crust.
9. _____________ & ____________ are the few examples of Extrusive / Volcanic (fine-grained) and Intrusive / Plutonic (coarse-grained) rocks respectively.
10. The rocks containing more than ________ of silica are termed as ___________ and those with large amount of basic components & _____________ of silica are termed as ______________.
11. A coarse-textured light coloured rock that containing 60 – 70 % feldspar is ___________ and the fine-grained rock with the above comparable composition is ______________.
12. A fine-textured and dark coloured rock which contains 50 % Feldspar and 50 % ferromagnesian minerals is _______________ and the coarse-grained rock with the comparable composition is _____________.
13. Diagenesis refers to ____.
14. Marble, a Metamorphic rock is derived from the Sedimentary rock ________________.
15. _______________ is an example of finely foliated / laminated metamorphic rock, while _____________ is an example of non-foliated, crystalline metamorphic rock.
16. Petrography which deals with the ____.
17. Petrogenesis which is the study of ____.
18. A rock, chiefly composed of Calcite mineral is __________________.
19. The banded / laminated character is the most peculiar feature of _______________ rocks.
20. Due to Thermal Metamorphism, Granite, a coarse-grained rock is converted into __________.
21. In Dynamothermal Metamorphism, ___________ is the dominant factor which brings changes.
22. Based on Texture and Structure of minerals, the Metamorphic rocks are divided into ___________, ____________ and ____________.
23. _______________ is the most common feature of Sedimentary rocks and as such are also termed as ________________.
24. ____________ is an example of Sedimentary rock in which Quartz is the predominant mineral.
25. ________is an example of fine-grained Detrital rock, composed of clay and silt sized particles.

Differentiate the following

1. Igneous rocks Vs Metamorphic rocks
2. Petrography Vs Petrogenesis
3. Mono-mineralic rocks Vs poly-mineralic rocks
4. Intrusive rocks Vs Extrusive rocks
5. Volcanic rocks Vs Plutonic rocks
6. Over-saturated rocks Vs Under-saturated rocks or Acid Igneous rocks Vs Basic Igneous rocks
7. Granite Vs Basalt
8. Sandstone Vs Limestone
9. Gneiss Vs Schist
10. Primary rocks Vs Secondary rocks

Write a short note on the following

1. Igneous rocks / Sedimentary rocks / Metamorphic rocks
2. Definition of rock
3. Petrology
4. Processes involved in formation of rocks
5. Rock cycle
6. Classification of rocks (based on mode of origin)
7. Classification of igneous rocks (based on mode of origin)
8. Classification of igneous rocks (based on SiO_2 content / chemical composition)
9. Stratification
10. Different stages of formation of Sedimentary rocks
11. Diagenesis
12. Classification of Sedimentary rocks
13. Metamorphism and its types
14. Classification Metamorphic rocks
15. Pre-existing and their equivalent metamorphic rocks

Answers - Soil Forming Rocks

Fill in the blanks

1. Pumice
2. Igneous – Primary / Massive / Crystalline / Magmatic, Sedimentary – Secondary / Stratified / Clastic / Aqueous , Metamorphic – Metamorphosed / Changed rock
3. Acid Igneous – Granite, Basic Igneous – Basalt
4. Magma is defined as a complex hot solution of silicates containing water vapour ands gases having a temperature ranging from 700 – 1400 C and originating at great depths in the earth's crust.
5. When the molten mass of Magma flows out on the surface of the Earth, it is called as Lava.
6. Olivinite, Dunite
7. Coarse grained – Granite, Fine-grained – Basalt
8. Igneous
9. Extrusive / Volcanic (fine-grained) – Basalt / Rhyolite, Intrusive / Plutonic (coarse-grained) – Granite / Gabbro
10. 60 % , Acid / Over-saturated rocks, 41 – 50 % , Basic / Under-saturated rocks
11. Granite, Rhyolite
12. Basalt, Gabbro
13. Transformation of unconsolidated sediments to hard rock
14. Limestone
15. Schist, Marble
16. Description of rocks
17. Origin of rocks
18. Limestone
19. Metamorphic
20. Gneiss
21. combination of pressure and temperature
22. foliated, unfoliated & granular
23. Stratification, Stratified rocks
24. Sandstone
25. Shale

III. Soil Forming Minerals

Fill in the blanks

1. The examples of minerals consisting of only one of the naturally occurring chemical elements__________________________________.
2. ____________________ is an example of orthosilicates / single chained Inosilicates / double chained Inosilicates / Phyllosilicates / Tectosilicates.
3. _______________ is the fundaments building block of all the silicate minerals of the earth's crust.
4. _____________________ are the few examples of ferromagnesian minerals.
5. Black mica is otherwise called as ___________ while white mica is otherwise referred as __________.
6. Minerals that have persisted with little change since they were extruded from molten lava are known as ________________________.
7. ________________ is the most abundant / dominant mineral in the earth's crust.
8. The example of the mineral that is olive-green in colour.
9. _____________________ are the few examples of heavy / light minerals.
10. ___________________ is considered as the non-ferromagnesian mica / phyllosilicates mineral.
11. The examples for the amorphous minerals are _________________________.
12. __________________ are the few examples of dark / light coloured minerals.
13. ______________ is the softest mineral known and _____________ is the hardest mineral known.
14. __________________ is the mineral that is called as "fools gold"
15. __________________ is the mineral which is called as Bog iron.
16. The mineral consisting of carbonates of calcium and magnesium is _______________.
17. ________ is the mineral which is the principal component of the common sedimentary rock, limestone.
18. On hydration, the mineral Haematite is converted into _______________.
19. Dolomite is less readily __________________ than Calcite.
20. ___________, ____________, ____________ and _________________ is an example of 1:1, 2:1 Expanding, 2:1 Non-expanding & 2.2 clay minerals.

Differentiate the following

1. Primary minerals Vs Secondary minerals
2. Essential Minerals Vs Accessory minerals
3. Light Minerals Vs Heavy minerals
4. Silicate Minerals Vs Non-silicate minerals
5. Ortho silicates Vs Inosilicates (Single and double chained)
6. Phyllosilicates / Layer Silicates Vs Tectosilicates
7. Ferromagnesian minerals Vs Non- ferromagnesian minerals
8. Soil Profile Vs Soil horizons
9. Biotite Vs Muscovite
10. Crystalline minerals Vs Non-crystalline minerals

Write a short note on the following

1. Definition of Minerals
2. Soil Profile
3. Relative abundance of rock-forming minerals
4. Different types of feldspars
5. Sequence of weathering stability of minerals
6. Types of layer silicate clay minerals
7. Classification of minerals
8. Physical properties of minerals
9. Quartz
10. Feldspars

Answers - Soil Forming Minerals

Fill in the blanks

1. Metal – Cu, Fe, Ca, gold, silver, Ni / Non-metal – C (graphite), S, Si
2. Ortho – Olivine, Ino (SS) – pyroxene, Ino (D) – Amphiboles, Phyllo – Micas, Tecto – Feldspar, Quartz
3. Silicate tetrahedron / silica tetrahedron
4. Olivine, Pyroxene, Amphiboles, Biotite
5. Biotite, Muscovite
6. Primary Minerals
7. Feldspars
8. Olivine
9. Heavy minerals (S.G > 2.85) – Haematite (5.3), Limonite (3.8), Pyrite (5.0), Olivine (3.5)
   Light minerals (S.G <2.85) – Quartz (2.60), Feldspar (2.65), Mica (2.5 – 2.75)
10. Muscovite (white mica)
11. Allophane, Chalcedony
12. Dark Coloured – Olivine (Olive green), Pyroxene (Dark green), Biotite (Black)
    Light Coloured – Muscovite (White), Anthracite (White), Quartz (Colourless)
13. Talc (Mho's harness scale – 1), Diamond (Mho's hardness scale – 10)
14. Pyrite (FeS_2)
15. Limonite ($2\ Fe_2O_3.\ 3\ H_2O$)
16. Dolomite [$Ca\ Mg\ (CO_3)_2$]
17. Calcite ($CaCo_3$)
18. Limonite ($2\ Fe_2O_3.\ 3\ H_2O$)
19. decomposed
20. 1 : 1 – Kaolinite, 2 : 1 Expanding – Montmorillonite, 2 : 1 Non-expanding – Illite, 2 : 2 / 2 : 1: 1 - Chlorite

Differentiate the following

1. **Primary Minerals Vs Secondary Minerals**

Primary Minerals	Secondary Minerals
❖ Minerals that have persisted with little change since they were extruded from molten lava are known as Primary Minerals ❖ The primary minerals are those which formed owing to the crystallization of the molten magma. ❖ A group of minerals such as Feldspar, Mica, Pyroxenes which occur originally in rock or derived from the weathering of rocks. ❖ Soils inherit primary minerals from their parent materials.	❖ The minerals formed due to weathering of pre-existing primary minerals are called as secondary minerals. ❖ The secondary minerals are formed at the earth's surface by weathering of the pre-existing primary minerals under variable conditions of temperature and pressure. ❖ The minerals that are formed by transformation from primary minerals or released by weathering. e.g. Clay minerals and oxides. ❖ Secondary minerals are the minerals formed in the environment from the primary minerals .

2. **Essential Minerals Vs Accessory Minerals**

Essential Minerals	Accessory Minerals
❖ The minerals which form the chief constituents of a rock and which are recognized as the characteristic components of that rock are classified as essential minerals. ❖ They from the major part of the rocks, and are instrumental in giving the rock its name. ❖ Occur in quantities varying from 95 – 98 %. ❖ e.g. Feldspar, Pyroxenes, Amphiboles, Micas, Calcite, Silicate minerals.	❖ The minerals which occur only in small quantities and whose presence or absence is of no consequence as far as the character of the rock is concerned are called accessory minerals. ❖ If the parent materials are transported and deposited by or in Ocean, they are called as Marine. ❖ Fine sediments deposited at the bottom of the sea and get exposed at the surface due to change in sea level. ❖ e.g. Apatite, Pyrite, Magnetite, Zircon, Topaz, Tourmaline.

3. **Light Minerals Vv Heavy Minerals**

Light Minerals	Heavy Minerals
❖ Light minerals are the minerals having the specific gravity below 2.85. ❖ e.g Quartz (2.60), Feldspar (2.65), Mica (2.5 – 2.75).	❖ Heavy minerals are the minerals having the specific gravity above 2.85. ❖ e.g. Haematite (5.3), Limonite (3.8), Pyrite (5.0), Olivine (3.5).

4. **Silicate Minerals Vs Non-silicate minerals**

Silicate Minerals	Non-silicate Minerals
❖ Silicate minerals are the minerals having silica tetrahedron as fundamental building block of their structure. ❖ Most dominant minerals (compounds containing silicon and oxygen, and one or more metal cations) in the earth's crust. ❖ e.g. Primary minerals - Quartz, Feldspar, Micas Secondary minerals – Clay minerals.	❖ Non-silicate minerals are the minerals having the compounds other than silica tetrahedron as fundamental building block of their geometric structure. ❖ Least dominant when compared to silicate minerals. ❖ May be oxides / hydroxides / carbonates / sulphates / phosphates. ❖ e.g Haematite, Limonite, Dolomite, Calcite, Pyrite, Apatite.

5. **Orthosilicates Vs Inosilicates**

Orthosilicates	Inosilicates
❖ The silicates that are made up of individual / independent tetrahedra alternating with positively charged metal ions (Fe, Mg), are Orthosilicates, of which Olivine is a typical example. ❖ No inter-tetrahedral bonding in Orthosilicates. ❖ In Orthosilicates, the negative charge provided by the oxygen in any one tetrahedral unit of the chain remains eight and hence the negative charge imbalance remains four.	❖ The silica tetrahedral units join together forming long chains which may be single (single-chained) or double (double-chained). ❖ In single-chained Inosilicates, the negative charge provided by the oxygen in any one tetrahedral unit of the chain reduced from eight (orthosilciates) to six and hence the negative charge imbalance is reduced from four (orthosilciates) to two. ❖ In double-chained Inosilicates, the negative charge provided by the oxygen in any one tetrahedral unit of the chain reduced from eight (orthosilciates) to 5.5 and hence the negative charge imbalance for each silicon is reduced from four (orthosilciates) to one and half. ❖ e.g. Single chained – Pyroxenes, Double chained – Amphiboles.

6. **Phyllosilicates / Layer Silicates Vs Tectosilicates**

Phyllosilicates / Layer Silicates	Tectosilicates
❖ Phyllosilicates are an important group of soil-forming minerals and are represented by micas (Biotite, muscovite). ❖ They have sheet structure of tetrahedral where each silicon ion shares three oxygen ions with adjacent silicon ion to from a pattern, like honey-comb. ❖ The basic structural unit of phyllosilicates is essentially formed by the condensation of two sheets of silicon-tetrahedra with one sheet of aluminium or magnesium octahedron.	❖ In Tectosilicates, the maximum linkage is attained in a 3-diamentioanl network in which all oxygen ions are shared between tetrahedra. ❖ Here, each silicon is associated with 4 shared oxygen ions. ❖ This results in four negative and positive charges for any tetrahedral unit, thereby achieving an electrically neutral structure. ❖ Typical examples are Feldspars and Quartz.

❖ In Phyllosilicates, the negative charge provided by the oxygen in any one tetrahedral unit reduced from eight (orthosilciates) to five and hence the overall is reduced from four (orthosilciates) to one only.	❖ IN its structure, all the four oxygen of the silica tetrahedron are shared by the neighboring silicon-tetrahedra. ❖ It means that there are two ions of oxygen for every ion of silicon thus forming a 3-diamentional framework.

7. **Ferromagnesian Minerals Vs Non- ferromagnesian Minerals**

Ferromagnesian Minerals	Non-ferromagnesian Minerals
❖ Ferromagnesian minerals are the minerals that are containing Fe and Mg in their structure. ❖ Here, the silicon-oxygen tetrahedral units are joined by ions of Fe and Mg. ❖ Typical examples are Olivine, Amphiboles, Pyroxenes, Biotite.	❖ Non-ferromagnesian minerals are the minerals that are devoid of Fe and Mg in their structure. ❖ Typical examples are Muscovite, Feldspars & Quartz.

8. **Soil Profile Vs Soil horizons**

Soil Profile	Soil horizons
❖ A vertical section of the soil through all its horizons and extending into the C horizon. ❖ It represents the succession of soil horizons down to and including the upper part of the undifferentiated parent material. ❖ The vertical section of the soil body above the parent material is called soil. ❖ The vertical section of the soil to expose the layering is called a profile.	❖ A relatively uniform layer that extends laterally (continuously or discontinuously) throughout the pedon; runs approximately parallel to the surface of the ground and differs from the related horizons in many chemical, physical and biological properties. ❖ A soil horizon a layer of soil that is approximately parallel to the soil surface and has properties that are produced by soil-forming processes.

9. **Biotite Vs Muscovite**

Biotite	Muscovite
❖ Biotite is Black mica. ❖ Black to dark green in colour ❖ It is a ferromagnesian mineral... ❖ Common in Igneous and Metamorphic rocks. ❖ One of the important mineral in the sand and silt fraction of most soils. ❖ Biotite weathers more rapidly than Muscovite.	❖ Muscovite is white mica. ❖ Muscovite is white in colour / colourless. ❖ It is a non-ferromagnesian mineral. ❖ Common in Igneous, Sedimentary and Metamorphic rocks. ❖ Muscovite is relatively resistant to weathering than Biotite.

10. **Crystalline minerals Vs Non-crystalline minerals**

Crystalline minerals	Non-crystalline minerals
❖ The minerals that are having crystalline structure are called as Crystalline minerals. ❖ The atoms in the minerals are having specific arrangement. ❖ Most of the soil-forming minerals are crystalline in nature.	❖ The minerals that are devoid of crystalline form in their structure are called as Non-crystalline minerals. ❖ The atoms in the minerals are not having specific arrangement and mostly in powder form. ❖ Typical examples are Allophane, Chalcedony that are predominant in young soils formed in Volcanic ash.

Write a short note on the following

1. **Minerals**
   - ❖ Minerals are solid substances and composed of atoms having an orderly and regular arrangement. (or)
   - ❖ Mineral is a naturally occurring homogenous element or inorganic compound that has a definite chemical composition and a characteristic geometric form. (or)
   - ❖ Mineral is an inorganically formed solid composed of specific elements in a structural arrangement characteristic of the mineral

2. **Soil profile**
   - ❖ A vertical section of the soil through all its horizons and extending into the C horizon.
   - ❖ It represents the succession of soil horizons down to and including the upper part of the undifferentiated parent material.
   - ❖ The vertical section of the soil body above the parent material is called soil solum, and that including the parent material but not parent rock, if any, is the soil profile.

3. **Relative abundance of rock-forming minerals**

Minerals (arranged in order of their crystallization)	**Per cent distribution**
A. Primary minerals	
Ferromagnesians	
• ***Ortho or Inosilicates*** ***Olivine, Pyroxenes, Amphiboles, etc.***	16.8
• ***Phyllosilicates*** ***Biotite and Muscovite***	3.6
Non-ferromagnesians	
• ***Tectosilicates*** ***Feldspars*** ***Quartz***	61.0 11.6
B. Secondary Clay minerals	1.0
C. Others	6.0
TOTAL	100.0

4. **Different Types of Feldspars**
   - ❖ Calcium Feldspar – Anorthite Ca Al_2 Si_2O_8)
   - ❖ Sodium Feldspar – Albite (Na Al Si_3O_8)
   - ❖ Potash feldspar – Orthoclase (K Al Si_3O_8) / Microcline (K Al Si_3O_8)

5. **Sequence of weathering stability of minerals**
   - ❖ Quartz > Feldspar > Mica > Ferromagnesian minerals
   - ❖ Quartz (most resistant and stable) > Muscovite , K Feldspar > Na & CA Feldspar > Biotite, Hornblende, Augite > Amphiboles > Pyroxenes > Olivine > Dolomite & Calcite > Gypsum (least resistant and least stable)

6. **Types of Layer Silicate Clay Minerals**
   - ❖ 1 : 1Lattice type – Kaolin group - Kaolinite
   - ❖ 2 : 1 Lattice type Expanding – Smectitic group – Smectite, Montmorillonite
   - ❖ 2 : 1 Lattice type Non-expanding – Mica group – Illite
   - ❖ 2 : 2 / 2 : 1: 1 mixed layer type - Chlorite

7. **Classification of Minerals**

Based on Quantity	Based on Mode of Origin	Based on Specific Gravity	Based on Chemical Composition	Based on Colour	Based on presence of Silicates
Essential Minerals – Form major part of the rocks and occur in quantities from 95 – 98 % e.g. Calcite, Silicate minerals	**Primary Minerals –** formed from crystallization of magma e.g. micas, quartz,	**Light Minerals –** Having Sp.Gr. < 2.85 e.g. Quartz, Feldspar, Mica	**Native Elements –** e.g. Graphite, Sulphur, Gold, Copper	**Light Coloured Minerals –** e.g. Muscovite (White), Anthracite (White), Quartz (Colourless)	**Silicates –** e.g. Micas, Olivine, Feldspars
Accessory Minerals – Occur in subsidiary amount varying from 2 – 5 %. e.g Pyrite, Apatite, Magnetite	**Secondary Minerals –** formed due to decomposition or alteration of primary minerals e.g. clay minerals	**Heavy Minerals –** Having Sp.Gr. > 2.85 e.g. Pyrite, Limonite, haematite, Olivine	**Oxides and Hydroxides –** e.g. Quartz, Haematite, Limonite	**Dark Coloured Minerals –** Olivine (Olive green), Pyroxene (Dark green), Biotite (Black)	**Non-silicates –** e.g. Gypsum, Pyrite, Calcite, Dolomite, etc.,
			Sulphates – e.g. Gypsum		
			Sulphides – e.g. Pyrite		
			Carbonates – e.g. Calcite, Dolomite		
			Halides – e.g. rock salt		
			Silicates – e.g. Micas, Olivine, Feldspars		

8. **Physical Properties of Minerals**

- ❖ Colour, Streak, Lusture, Striation, Hardness, Transparency, Specific gravity, Tenacity, Fracture & Cleavage, State of Aggregation / crystal from, Taste, Magnetism, Effervescence, Birefringence, Fluorescence / Double refraction.

9. **Quartz**

- ❖ One form of silica and having the formulae of SiO_2
- ❖ One of the important primary minerals that is abundant in rocks and soils.
- ❖ Usually dominant minerals of fine sand and coarse silt fraction of soil.
- ❖ One of the important examples of Tectosilicates.
- ❖ Most resistant mineral to physical and chemical weathering, as the structure is densely packed.
- ❖ The abundance of Quartz in most soils is next to Feldspar.

10. **Feldspar**

- ❖ Most dominant primary mineral in soils.
- ❖ Most abundant rock-forming mineral and constitute nearly 61 % of the minerals that are found in the earth's crust.
- ❖ One of the important Non-ferromagnesian minerals
- ❖ Act as store house of Na, Ca, K and many trace elements in soils.

IV. Weathering of Rocks and Minerals

Fill in the blanks

1. Minerals that have persisted with little change since they were extruded from molten lava are known as _______________________.
2. Regolith refers to ___________________________________.
3. The mineral that is most resistant to weathering is / subjected to crystallization at last is ___________________.
4. The mineral that, is most susceptible to weathering is / subjected to crystallization at first / weathers more rapidly ___________________.
5. Biotite weather more _______________ than Muscovite.
6. Coarse-textured (porous) sandstone will weather more ________________ than fine-textured (almost solid) basalt.
7. Dark-coloured minerals/rocks are subject to ___________ changes in temperature than light- coloured minerals / rocks.
8. Dark- coloured minerals/rocks are subject to weathering at a __________ rate than light-colored minerals / rocks.
9. Quartz responds far ____________ to chemical attack / decomposition than does a mineral, like olivine / pyroxene.
10. The oxidation-reduction processes are common in minerals containing Fe, Mn and S.
11. The most resistant products of weathering, such as ____________________________________, tend to accumulate in soil.
12. The rate of chemical weathering is much more __________________ in tropical areas (humid and warm) than in arid and hot desert areas / temperate areas.
13. The fine-grained feldspar decomposes more _______________ than coarse-grained feldspars when exposed to percolating water.
14. Sand-sized quartz is extremely ____________________ to chemical weathering as compared with clay-size quartz.
15. Oxidation occurs in well aerated rocks containing _____________________ minerals.
16. Hydration results in _____________ in volume of minerals.
17. Solum consists of ____________________ horizons.
18. Solum is the outcome of _______________ processes.
19. The order of weathering of different feldspars is _____________________________.
20. The dark-coloured ferromagnesian minerals are more __________ to chemical weathering than feldspars and quartz.

Differentiate the following

1. Disintegration / Physical Weathering Vs Decomposition / Chemical Weathering
2. Regolith Vs Bed rock
3. Regolith Vs True Soil / Solum
4. Hydrolysis Vs Hydration
5. Oxidation Vs Reduction

Write a short note on the following

1. Carbonation
2. Exfoliation / Onion Type weathering / Thermal Weathering
3. Weathering Sequence
4. Weathering Indices
5. Solution / Dissolution

Describe the following

1. Explain in detail the various weathering processes / different types of weathering of rocks and minerals. (or)
   Discuss in brief about the physical weathering of rocks and minerals. (or)
   Define weathering of rocks and discuss the process of physical weathering of rocks and minerals.

Answers - Weathering of Rocks and Minerals

Fill in the blanks

1. Primary minerals
2. Unconsolidated residues of the weathering rock on the earth's surface (A + B + C)
3. Quartz
4. Olivine
5. easily
6. readily
7. fast
8. faster
9. slowly
10. Fe, Mn and S
11. hydrous oxides of iron and aluminium
12. Pronounced
13. rapidly
14. resistant
15. ferromagnesian
16. increase
17. A + B
18. Pedogenic
19. K Feldspar, Na Feldspar and Ca Feldspar
20. more susceptible

Differentiate the following

1. **Disintegration / Physical Weathering Vs Decomposition / Chemical Weathering**

Disintegration / Physical Weathering	Decomposition / Chemical Weathering
❖ It is otherwise called as physical / mechanical weathering ❖ By disintegration, the rocks are disintegrated, that is broken down into comparatively smaller pieces, without producing any new substances. ❖ The result of physical weathering can be compared with hammering or crushing of the rocks under great force. ❖ The different agencies that take part I the process of Physical weathering are physical condition of rocks, change in temperature, action of water, action of wind, atmospheric electric phenomena.	❖ It is otherwise called as chemical weathering. ❖ It is more complex in nature and involves the transformation of original material into some new compounds (not present in original rock), that is, it brings about alteration in minerals. ❖ The chemical weathering of Feldspar, for example, produces the clay mineral, which has a different composition ands physical properties. ❖ The chemical reactions involved in decomposition are hydration, hydrolysis, solution, carbonation, oxidation & reduction.

2. **Regolith Vs Bedrock**

Regolith	Bedrock
❖ Regolith is unconsolidated residues of the weathering rock on the earth's surface or above the solid rock. ❖ The Regolith, or at least its upper portion, may therefore be termed as parent material. ❖ Bedrock is converted into Regolith through the process of weathering. ❖ Bedrock (R) → Regolith (C) → solum (A + B) ❖ It includes A + B + C horizons.	❖ Bedrock is the consolidated mass of rock at the surface of the earth or at some depth beneath the surface of the earth. ❖ Originally the whole surface of the earth passed through a molten stage and the first solid rock was derived from this molten material, known as 'Magma". ❖ It is designated as 'R'. ❖ Rocks which are the original starting point in the weathering process which lead to the formation of soil.

3. **Regolith Vs Solum / True Soil**

Regolith	Solum / True Soil
❖ Regolith is a loose, unconsolidated residues of the weathering rock on the earth's surface or above the solid rock. ❖ It is the result of disintegration and decomposition of the exposed bedrock. ❖ It is also termed as parent material of soil. ❖ The Regolith, or at least its upper portion, may therefore be termed as parent material. ❖ It may be a few meters to several meters in thickness. ❖ It may be derived from the underlying bedrock or may have been transported by the action of water, wind or ice. ❖ Its upper 2 – 3 meters, being nearer to the atmosphere and subjected to weathering and various soil-forming processes, differs from the Regolith below. ❖ Bedrock is converted into Regolith through the process of weathering. ❖ Bed rock (R) → Regolith (C) → solum (A + B) ❖ It includes A + B + C horizons.	❖ It refers to the upper and most weathered part of soil profile. ❖ The upper and biochemically weathered part of Regolith, considered as the true soil, differs from the material below by having high organic matter content, abundance of roots, and more intense weathering, and distinct horizonation / layering, resulting from different soil-forming factors and processes. ❖ Soil-forming factors and processes interplay to form true soil from Regolith. ❖ There are two consecutive but overlapping stages leading to the formation of true soil. ❖ a. the weathering of rock into Regolith (C) b. the formation of true soil / solum (A + B) from the Regolith (C). ❖ It includes A + B Horizons, sometimes O + A + B horizons.

4. **Hydrolysis Vs Hydration**

Hydrolysis	Hydration
❖ One of the most important processes of chemical weathering. ❖ It is mainly due to the property of dissociation of water into H^+ and OH^- ions which chemically combine with minerals and bring about changes, such as exchange, decomposition of crystalline structure and formation of new compounds. ❖ In rocks / soils, the hydrolysis is very conveniently brought about by ionized water when it comes in contact with rock-forming or silicate minerals. ❖ e.g. $KAlSi_3O_8$ (Feldspar) + HOH → $HAlSi_3O_8$ (acid silicate clay) + KOH ❖ $HAlSi_3O_8$ + HOH → $Al_2O_3 . 3H_2O$ (Gibbsite) + H_2SiO_3 (silicic acid) ❖ It provides clay, bases and silicic acid that are available to plants. ❖ Water, often containing Co_2 reacts with the minerals directly to produce clay minerals, cations, anions, soluble silica which are all made available to plants.	❖ It is the chemical union of water with a substance leading to a change in structure of that substance. ❖ It is also called as chemical combination of water or addition of water of crystallization. ❖ Due to hydration, there is swelling and increase in volume or bulk of the minerals. ❖ The minerals thereby lose their lusture and become soft. ❖ It is one of the most common processes in nature and works with secondary minerals such as aluminium oxide, iron oxide, gypsum, whereby these minerals absorb water, expand and tend to fall apart. ❖ $2Fe_2O_3 + 3H_2O \rightarrow 2Fe_2O_3 . 3H_2O$ (Haematite) (Limonite) ❖ $Al_2O_3 + 3H_2O \rightarrow Al_2O_3 . 3H_2O$ (Haematite) (Gibbsite) ❖ $CaSo_4 + 2H_2O \rightarrow CaSO_4 . 2H_2O$ (Anhydrite) (Gypsum) ❖ On dehydration, the mineral is converted back to its original form due to lose of water of crystallization.

5. **Oxidation Vs Reduction**

Oxidation	Reduction
❖ Oxidation is the absorption of an oxygen ion by the mineral usually from oxygen dissolved in water. ❖ The minerals combine / react with (atmospheric) O_2 and the process of combination is known as oxidation. ❖ Water takes oxygen of atmosphere in the solution form and penetrates to every cracks ad crevices of rocks and thus brings air into intimate contact with minerals. ❖ $4 FeO + O_2 \rightarrow 2 Fe_2O_3$ (Haematite) ❖ $4Fe_3O_4 + O_2 \rightarrow 6 Fe_2O_3$ (Magnetite) (Haematite) ❖ $2 Fe_2O_3 + 3 H_2O \rightarrow 2 Fe_2O_3. 3H_2O$ (Haematite) (Limonite) ❖ The oxidation is more active in the presence of water and the ferromagnesian minerals.	❖ The process of removal or loss of oxygen is called as reduction. ❖ It is the reverse of oxidation and the chemical removal of oxygen results in reduction. ❖ It is equally important in changing soil colour to grey, green, blue as ferric ions are converted into ferrous ion compounds. ❖ Under the conditions of excess water or water-logged condition (less / no oxygen), reduction takes place. ❖ Under reduction the mineral Haematite is reduced to ferrous oxide as follows ❖ $2 Fe_2O_3$ (Haematite) $\rightarrow 4 FeO$ (Ferrous oxide) $+ O_2$ ❖ The reduction is more active in the absence of oxygen and ferromagnesian minerals.

<u>Write a short note on the following</u>

1. **Carbonation**
   - ❖ Carbonation is the combination of CO_2 with any base and this process is quite effective in decomposing the minerals of rocks and soils.
   - ❖ The rate of dissolution and decomposition of minerals increases on addition of organic matter due to the production of CO_2 and organic acids.
   - ❖ The CO_2 also produced by plant roots, gets readily combines with bases to produce carbonates and bicarbonates as follows:
   - ❖ $CO_2 + 2 KOH \rightarrow K_2CO_3 + H_2O$
     $K_2CO_3 + H_2O + CO_2 \rightarrow 2 KHCO_3$
   - ❖ The itching effect of carbonated water upon some rocks, especially Limestone and those containing cementing substances (sandstone, Conglomerates), leads to removal of cementing substances and thereby disintegration.
   - ❖ Through the process of carbonation, the insoluble calcium carbonate is converted into soluble calcium bicarbonate as follows:
   - ❖ $CO_2 + H_2O \rightarrow H_2CO_3$
     CaCO3 (insoluble) + $H_2CO_3 \rightarrow Ca(HCO_3)_2$ (soluble)
2. **Exfoliation / Onion Type Weathering / Thermal weathering**
   - ❖ The variation of temperature, especially if quite wide and sudden, greatly influences the disintegration of rocks (thermal weathering).
   - ❖ The temperature of a rock at its surface is very different from that of the part beneath. Moreover the minerals within a rock also vary in their rate of expansion and contraction.
   - ❖ The differential expansion and contraction of minerals in a rock surface, due to variation in temperature, leads to generate stress between the heated (expanded) surfaces and the cooled unexpanded parts, resulting in fragmentation of rocks.
   - ❖ This process, with time, may cause the surface layer to peel off from the parent mass and the rock may ultimately disintegrate and this phenomenon is called as Exfoliation.
   - ❖ The peeling off of the plates from the parent rock due to differential expansion and contraction of minerals as a result of changes in temperature is popularly known as Onion-type weathering.

3. **Weathering Sequence**

- The minerals can be arranged in the order of stability (resistance) or weatherability. In view of differences in surface area and consequent reactivity, it is desirable to separate the mineral particles into two classes viz.clay-sized and sand-silt-sized.
- The generalized order of weathering-resistance of the sand and silt-sized minerals is as follows:
- Quartz (most resistant and stable) > Muscovite , K Feldspar > Na & CA Feldspar > Biotite, Hornblende, Augite > Amphiboles > Pyroxenes > Olivine > Dolomite & Calcite > Gypsum (least resistant and least stable)
- It is comparable to the Goldich Weathering / Stability Sequence of 1938 and also follows Bowen's fractional crystallization series.
- The above sequence is in the increasing order of weathering and crystallization and is in the decreasing order of resistance to weathering. It shows that the Gypsum is the first (crystallization at early stages at high temperature) and Quartz, the last mineral to crystallize. Thus, higher the temperature of crystallization of a mineral from the magma, the more unstable it is.
- The order may change slightly depending on the climatic and environmental conditions.
- This explains the dominance of Gypsum, Calcite in the Arid zones and their absence in the soils of humid temperate regions where Quartz, Muscovite, and K-feldspar dominate in the coarse fraction of most soils.

4. **Weathering Indices**

- Different combination of factors of weathering may produce a given degree or stage of weathering.
- The stage of weathering is described in terms of weathering index / indices.
- The weathering sequence of Jackson and associates, as slightly modified, gives weathering stability index numbers to clay-sized minerals.
- Increasing index numbers indicate increasing relative resistance to weathering.
- The increasing order of weathering stability and weathering index numbers (given in parenthesis) for the clay-sized minerals is as follows:
- Gypsum, NaCl (1) > Calcite, Dolomite, Apatite (2) > Olivine – Hornblende, Pyroxenes, Diopside (3) > Biotite (4) > Albite (5) > Quartz (6) > Illite (7) > Vermiculite (8) > Montmorillonite (9) > Kaolinite (10) > Gibbsite, Allophane (11) > Haematite, Geothite, Limonite (12) > Anatase, Zircon, Rutile, Ilmenite, Corundum (13)

5. **Solution / Dissolution**

- Some substances like NaCl and KCl in a rock are directly soluble in water. When the soluble substances are removed by the continuous action of flowing or percolating water, the rock no longer remains solid and develops holes, rills or rough surface and ultimately decomposes and such process of chemical decomposition is called as Solution.
- The action is considerably increased when the water is acidified by the dissolution of organic and inorganic acids. Such waters may dissolve not only alkaline earths but also silica.
- e.g. $NaCl + H_2O \rightarrow Na^+ + Cl^- + H_2O$ (dissolved ions with water)

V. Factors of Soil Formation / Soil Forming Factors

Fill in the blanks

1. The soil formation is a process of two consecutive, but overlapping stages ____________________ and __________________.
2. Dokuchaev (1889) put forward that the soil develop as a result of the interplay of soil forming factors, *viz* ______________, _______________ and ______________.
3. Jenny (1941) formulated that the properties of soil is a function of ___________________.
4. The parent materials that are transported by river is called as ___________________.
5. Alluvium / Colluvium / Lacustrine / Loess (Aeolian) / Dunes / Marine / Moraine (till) refers to ____.
6. The role of parent material / relief / time on soil formation is _____________________.
7. The role of climate and biosphere on soil formation is ________________________.
8. Basalt, an igneous rock, may gives rise to highly weathered ________________ soils in the humid tropics, or _____________ soils in the semi-arid/sub-humid tropics.
9. The silica : sequioxide ratio of least weathered alluvial soils / Aeolian-derived soils / soils of arid region is ____________ and __________________ in highly weathered soils of the tropical region.
10. The soils on steep slopes are generally ____________ in soil depth and have _____________ soil profile with _____________________ horizonation.
11. The direction of the slope is called as _____________________.
12. The southern exposures are __________________ then northern exposures.
13. The soil catena concept was developed by _______________________.
14. The time that nature devotes to the formation of soils is termed as _______________.
15. ____________ is considered as a genetic factor that locally influences soil formation.
16. Amongst all the soil forming factors, ____________ is perhaps the most influential in soil development.
17. The resultant end product of soil forming factors is ___________.
18. Acid igneous rocks (like granite, rhylolite) produce ____________ textured soils while the basic igneous rocks (like basalt) produce ___________ textured soils.
19. Highly siliceous parent materials, such as sandstone tend to give ____________ soils, while basic rocks and fine-grained sediments tend to develop ____________ textured soils.
20. The rate of profile development is ___________ in warm humid climate then cold humid climate or the rate of profile development / aging is ____________ in cold / hot / arid / temperate climate than the warm humid / tropical climate.

Differentiate the following

1. Active Soil Forming Factors Vs Passive Soil Forming Factors
2. Moraine Vs Marine
3. Alluvium VS Colluvium
4. Endodynamorphic soils Vs Ectodynamaorphic soils
5. Weathering Vs Soil formation

Write a short note on the following

1. SiO_2 / R_2O_3 ratio
2. Soil formation
3. Soil Forming Factors
4. Types of Transported Parent materials
5. Chronosequence
6. Lithosequence
7. Toposequence / Clinosequence
8. Soil Catena
9. Pedologic time
10. Weathering stages in soil formation

Describe the following

1. Explain in detail about the role of various soil forming factors on soil formation.
   Discuss in brief about the active soil forming factors. (or)
   Briefly discuss the passive soil forming factors.

Answers - Factors of Soil Formation / Soil Forming Factors

Fill in the blanks

1. the weathering of rock (R) into Regolith (C) and the formation of true soil / solum from the Regolith
2. parent material, climate and organisms
3. S = f (cl, b, r, p, t……)
4. Alluvium
5. Alluvium – parent material transported by river
   Colluvium - parent material transported by gravity
   Lacustrine - parent material transported by lake
   Loess / Aeolian - parent material (very fine silt and clay sized) transported by wind
   Dunes - parent material (coarse sand sized) transported by wind
   Marine - parent material (coarse sand sized) transported by Ocean
   Moraine / Till - parent material (coarse sand sized) transported by Ice / melting glaciers
6. Passive
7. Active
8. Red, Black cotton
9. > 2, < 2
10. shallow, weakly-developed, less distinct
11. Aspect
12. Warmer
13. Milne (1935)
14. Pedologic time
15. Relief
16. Climate
17. Soil / Solum
18. light-textured, fine-textured
19. very sandy (coarse), fine
20. Faster, Slower

Differentiate the following

1. **Active Soil Forming Factors Vs Passive Soil Forming Factors**

Active Soil-forming Factors	Passive Soil-forming factors
❖ Active soil-forming factors are those which supply energy that acts on the mass for the purpose of soil formation. ❖ These factors are Climate and Biosphere / Vegetation. ❖ The development of soil profile is mostly influenced by active soil-forming factors (Zonal soils).	❖ The passive soil-forming factors are those which represent the source of soil forming mass and conditions affecting it. ❖ They provide base on which the active soil forming factors work or act for the development of soil. ❖ These factors are parent material, relief / topography and time.

2. **Moraine Vs Marine**

Moraine	Marine
❖ If the parent materials are transported and deposited by or in ice, they are called as Moraine. ❖ They are otherwise called as Till. ❖ The materials are deposited by the melting glaciers. ❖ Big boulders in the Kangra valley of Himachal Pradesh are an example.	❖ If the parent materials are transported and deposited by or in Ocean, they are called as Marine. ❖ Fine sediments deposited at the bottom of the sea and get exposed at the surface due to change in sea level.

3. **Alluvium Vs Colluvium**

Alluvium	Colluvium
❖ If the parent materials are transported and deposited by or in river (flood-plain, terraces), they are called as Alluvium. ❖ Deposits are carried in suspension by rivers. ❖ Most fertile soils in the world are developed on alluvium.	❖ If the parent materials are transported and deposited by or in gravity, they are called as Colluvium. ❖ The stratified materials are deposited on the foot slopes due to soil creep or flow.

4. **Endodynamorphic soils Vs Endodynamorphic soils**

Endodynamorphic soils	Endodynamorphic soils
❖ There are soils wherein the composition of parent material subdues the effect of climate and vegetation; such soils are called as Endodynamorphic soils. ❖ Such soils are temporary and persist only until the chemical decomposition becomes active under the influence of climate and vegetation. ❖ The soils with properties that have been influenced primarily by the parent materials.	❖ Ectodynamaorphic soils are the soils which develop a normal profile under the influence of climate and vegetation. ❖ The effect of climate and vegetation on soils is more predominant than the effect of parent materials on soils. ❖ The soils with properties that have been influenced by factors other than parent materials.

5. **Weathering Vs Soil Formation**

Weathering	Soil Formation
❖ First stage of soil formation ❖ Weathering is disintegration and decomposition of rocks giving rise to a loose and unconsolidated material, called Regolith. ❖ Rock (R) → Regolith (C) ❖ Destructive in nature.	❖ Second stage of soil formation ❖ Soil formation is the evolution of true soil from the Regolith / parent material takes place by the combined action of soil-forming factors and processes. ❖ Regolith (C) → True Soil / Solum (A + B) ❖ Constructive in nature.

Write a short note on the following

1. **SiO_2 / R_2O_3 ratio**
   - ❖ SiO_2 / R_2O_3 ratio is the ratio between silica and sequioxide (Fe and Al oxides)
   - ❖ Important index of weathering of rocks and minerals and in turn soil formation.
   - ❖ The ratio is maximum (> 2) in basic or least weathered alluvium and Aeolian-derived soils / soils of arid and semi-arid and temperate regions and minimum (< 2) in the soils of highly weathered tropical soils.
   - ❖ At constant high moisture content, the ration decreases as the temperature increases.
   - ❖ Arid region - > 3.5, cold - humid region – 3 to 3.5, sub-humid region – 2 to 3, tropical region - < 2.
2. **Soil Formation**
   - ❖ Second stage of soil formation
   - ❖ Soil formation is the evolution of true soil from the Regolith / parent material takes place by the combined action of soil-forming factors and processes.
   - ❖ Regolith (C) → True Soil / Solum (A + B)
   - ❖ As the soil forming factors are more effective in bringing about changes in any given soil, the soil formation is therefore considered a constructive process.

3. **Soil Forming Factors**
   - ❖ The soil-forming factors are the natural agencies that are responsible for the formation of soil.
   - ❖ The evolution of true soil from the Regolith takes place by the combined action of soil-forming factors and processes
   - ❖ They are grouped as active soil-forming factors (climate and vegetation / biosphere) and passive soil-forming factors (parent material, relief / topography and time)
   - ❖ The soil forming factors are interrelated and compliment each other to give rise to the true soil / solum.
   - ❖ The passive soil-forming factors provide a base on which the active soil-forming factors work or act form the development of soil.

4. **Types of Transported parent materials**

Agent	Transported and deposited by or in	Name of the deposit or parent material
Water	River	Alluvium
	Lake	Lacustrine
	Ocean / Sea	Marine
Wind	Wind	Sand Dunes (Sand & Silt sized)
		Loess / Aeolian (Silt & Clay sized)
Gravity	Gravity	Colluvium
Ice	Ice / Glaciers	Till / Moraine

5. **Chronosequence**
   - ❖ A sequence of related soils that differ from one another primarily due to the variation in time (Pedologic time) as a soil-forming factor.

6. **Lithosequence**
   - ❖ A sequence of related soils that differ from one another in certain characteristics as a result of variations in parent material.

7. **Toposequence / Clinosequence**
   - ❖ A sequence of related soils that differ from one another in respect of topography as a soil-forming factor, is termed as Toposequence.
   - ❖ A sequence of related soils that differ from one another primarily because of the differences in degree of slope on which they are formed, is termed as Clinosequence.

8. **Soil Catena**
   - ❖ The Soil Catena concept was developed by Milne (1935)
   - ❖ A sequence of related soils developed from similar parent material under similar climatic conditions but varying conditions of relief and drainage.
   - ❖ An obsolete term in favour of Toposequence.
   - ❖ A Toposequence formed entirely in one kind of parent material is called as a soil catena (named from the Latin term meaning chain). A sequence of related soils that differ from one another in certain characteristics as a result of variations in parent material.
   - ❖ Where the parent material is similar throughout the Toposequence, the group of soils is called as a Catena.

9. **Pedologic Time**
   - ❖ The time that nature devotes to the formation of soil is termed as Pedologic Time.
   - ❖ The period taken by a given soil from the stage of weathered rock (Regolith) up to the stage of maturity is considered as Pedologic Time.
   - ❖ All soils do not age at the same rate.
   - ❖ It has been reported that it takes hundreds of years to develop an inch of soil.

10. **Weathering stages in soil formation**
   - ❖ Van Baren and Mohr suggested five stages of weathering that are dependant on mineralogical features of soils as follows.
   - ❖ **Initial –** unweathered parent material
   - ❖ **Juvenile –** Weathering started; but much of the original material still unweathered.
   - ❖ **Virile –** Easily weatherable mineral fairly decomposed; clay content increased; slowly weatherable minerals still appreciable.
   - ❖ **Senile –** Decomposition reaches at a final stage; only most resistant minerals survive.
   - ❖ **Final –** Soil development completed under prevailing conditions.

VI. Processes of Soil Formation / Soil Forming Processes

Fill in the blanks

1. The soils having higher oxidation-reduction potential are generally ___________than the soils with low oxidation-reduction potential.
2. Eluviation refers to ____.
3. The Process of Leaching refers to ____.
4. Illuviation refers to ____.
5. If the constituents are translocated in suspension, it is referred as ________________, while the constituents are translocated in solution, it is called as _____________.
6. Argillation refers to ____.
7. Laterization is otherwise called as ____.
8. __________ is an example of soil-forming process, which leads to the formation of Hydromorphic Soils.
9. _________ & __________ is analogous to Emigration / Washing- out and Immigration / Washing-in respectively.
10. The order of relative mobility of elements / soil constituents is _________________________.
11. The accumulation of calcium carbonate in lower layers of soil profile may results in the development of _____________ horizon and is designated as __________.
12. Calcification is the specific Pedogenic process of ________ & ___________ climatic region.
13. Differentiation of soil in different horizons along the depth of soil body is ______________.
14. The B horizon is sometimes referred as ______________.
15. _________ horizon is the zone of maximum accumulation of soil constituents such as clay, R_2O_3, SiO_2, etc., while the _________ horizon is the zone of leaching of soil constituents.
16. ______________ & _____________ are the few examples of soil-forming processes which lead to the formation of Zonal and Intrazonal soil respectively.
17. Matured soil profile will have _____________________ horizonation.
18. The process of Eluviation is highly predominant in _____________ soils.
19. The zone of loss is otherwise called as ___________, while the zone of accumulation is otherwise called as ________________.
20. The development of Podzols, Latosol is favoured by ___________ & __________ climatic conditions and ___________ & __________ parent materials respectively.
21. Podzol is the term which is of ___________ origin and means _________________________.
22. The fertility status of Podzols is low which is primarily due to _________________________.
23. The development of Glei horizon / Gley is mainly due to the Pedogenic process of _________.
24. Solonchak and Solonetz are the Russian terms which refer to _______ & __________ soils respectively.
25. Soil-forming processes are otherwise called as _____________________.

Differentiate the following

1. Basic / Fundamental Pedogenic Processes Vs Specific Pedogenic Processes
2. Eluviation Vs Illuviation
3. Calcification Vs Decalcification
4. Podzolization Vs Laterization / Ferritization / Latozation
5. Salinization Vs Desalinization
6. Alkalization / Solonization Vs Solodization / Dealkalization

Write a short note on the following

1. Humification
2. Gleization
3. Horizonation
4. Pedoturbation
5. Argillation
6. Lessivage

Describe the following

1. Briefly explain the various Pedogenic processes / Soil Forming Processes. or
   Give a brief account of soil-forming processes.
2. Define soil profile. Draw a sketch of an ideal soil profile indicating various horizons.

Answers – Processes of Soil Formation / Soil Forming Processes

Fill in the blanks

1. Fertile
2. the process of removal of constituents in suspension or solution by percolating water from the upper to lower layers
3. the movement and removal of material in solution from the soil.
4. the process of deposition of soil materials (removed from the eluvial horizon) in the lower layer
5. Eluviation, Leaching
6. the leaching of dispersed clay particles from upper to lower horizons
7. Ferritization / Latozation
8. Gleization
9. Eluviation, Illuviation
10. $Cl > SO_4 > Ca > Na > Mg > K > SiO_2 > R_2O_3$ ($Fe_2O_3 + Al_2O_3$)
11. Calcic, Bk
12. Arid & Semi-arid
13. Horizonation
14. Sub-soil
15. B, E
16. Zonal – Calcification, Podzolization, Laterization; Intrazonal – Gleization, Salinization, Alkalization / Solonization,
17. distinct
18. Podzols / Spodosols
19. Eluvial, Illuvial
20. Cool Humid, Warm Humid; Siliceous / Acidic / Coarse, Basic
21. Russian, Pod – Under, Zol - Ash like
22. Intensive leaching of soil constituents especially bases
23. Gleization
24. Saline, Alkali
25. Pedogeneic Processes

Differentiate the following

1. **Basic / Fundamental Pedogenic Processes Vs Specific Pedogenic Processes**

Basic / Fundamental Pedogenic Processes	Specific Pedogenic Processes
❖ They are the basic processes involved in soil formation. ❖ The fundamental Pedogenic processes include Humification, Eluviation, Illuviation, Horizonation. ❖ All these processes promote horizon differentiation. ❖ The above processes combine to result in a number of wide ranging soils that are observed on the surface of the earth.	❖ They are very specific pedogenic processes that are confined to specific climatic conditions / environments. ❖ The Specific pedogenic processes includes Podzolization, Laterization, Gypsification, Calcification, Decalcification, Salinization, Desalinization, Alkalization (Solonization), Dealkalization (Solodization), Gleization, Pedoturbation, Argillation, Lessivage. ❖ The basic Pedogenic processes provide a framework for later operation of more specific processes under specific environments.

2. **Eluviation Vs Illuviation**

Eluviation	Illuviation
❖ Eluviation is the process of removal of constituents in suspension or solution by the percolating water from the upper to lower layers. ❖ The Eluviation encompasses mobilization and translocation of mobile constituents resulting in	❖ If the parent materials are transported and deposited by or in gravity, they are called as Colluvium. ❖ The stratified materials are deposited on the foot slopes due to soil creep or flow.

textural differences. by percolation from upper to If the parent materials are transported and deposited by or in river (flood-plain, terraces), they are called as Alluvium. ❖ Deposits are carried in suspension by rivers. ❖ Most fertile soils in the world are developed on alluvium.	

3. **Calcification Vs Decalcification**

Calcification	Decalcification
❖ It is the process of precipitation and accumulation of $CaCO_3$ in some part of the profile. ❖ $CO_2 + H_2O \longrightarrow H_2CO_3$ ❖ $H_2CO_3 + Ca \longrightarrow Ca(HCO_3)_2$ (soluble) ❖ At ↑ T & ↓ CO_2, $Ca(HCO_3)_2 \longrightarrow CaCO_3(ppt) + H_2O + CO_2$ ❖ Accumulation of $CaCO_3$ may results in the development of Calcic horizon (Bk / Bca). ❖ Such $CaCO_3$-enriched Bk horizons are quite common in arid and semi-arid environments of Rajasthan, Punjab and Haryana where the parent materials are rich in lime. ❖ The soils with such a lime-enriched layer were termed by Marbut as Pedocals. ❖ Native vegetation varies from grass to desert shrubs.	❖ It is the reverse of Calcification. ❖ It is the process of removal of $CaCO_3$ or calcium ions from the soil by leaching. ❖ At ↓ T & ↑ CO_2, ❖ $CaCO_3 + H_2O + CO_2 \longrightarrow Ca(HCO_3)_2$ (soluble) (insoluble) ❖ In humid climate, the insoluble / $CaCO_3$ precipitate reacts with water containing dissolved CO_2 to from soluble calcium bicarbonate which may be completely leached out of the profile through high rainfall (percolating water).

4. **Podzolization Vs Laterization / Ferritization / Latozation**

Podzolization	Laterization / Ferritization / Latozation
❖ The Russain term 'Podzol' (Pod – under; Zola – ash-like) means ash-like horizon appearing beneath the surface horizon, as evidenced by the characteristic silica abundant ash-coloured grayish E horizon.	❖ The term 'Laterite' is derived from the word 'later' meaning brick or tile and refers to specifically a particular cemented horizon (massively impregnated with R_2O_3) in certain soils, when dried, become very hard like a brick.
❖ It is the process of soil formation resulting in formation of Podzols and Podzolic soils	❖ It is the process of soil formation which results in the formation of Laterites / Latosols / Oxisols.
❖ A cold, humid climate with high rainfall and low temperature throughout the year, is most favourable for Podzolization.	❖ The process operates most favourably in warm, humid climate with 2000 – 2500 mm rainfall and continuous high temperature (± 25 C) throughout the year.
❖ Siliceous / coarse sandy parent material, having poor reserves of weatherable minerasl, favours the process of Podzolization.	❖ Basic parent materials, having sufficient iron bearing ferromagnesian minerals which release Fe on weathering, are congenial for the development of Laterites.
❖ Acid-producing vegetation, such as Pines, Hemlock, Heath, Erica, Populus, etc. having less bases, are essential for the operation of this process.	❖ The rain forests of tropical areas, rich in bases, are favourable for this process
❖ It is type of acid leaching (eluviation) in which humus & R_2O_3 except silica, become mobile, leached out fro the upper layers and are redeposited in the lower layers.	❖ The high temperature, intense leaching and basic parent martial all favour the removal of silica (desilication) and accumulation of R_2O_3.

❖ Due to intensive leaching of bases, soil reaction is acidic in nature. ❖ Podzols are low in fertility due to intensive leaching of bases, humus and clay and are mainly used for forestry and / pastures, some are cultivated for crops, like oats, potatoes and clover. ❖ A typical and mature Podzol profile with distinct horizonation (A & B horizons show strong textural contrast with a bleached E horizon).	❖ Soil pH goes up to neutrality initially and acidic later on due to the removal of bases. ❖ Low in soil fertility and productivity due to intensive leaching of bases, low CEC, fixation of P, crusting and are mainly used for shifting agriculture, low intensity grazing, plantation of Coconut, Pine apple, Coffee, Banana and Arecanut. ❖ Though considerable eluviation takes place, there is no marked horizonation as the eluviated materials are not redeposited in the lower layers.

5. **Salinization Vs Desalinization**

Salinization	Desalinization
❖ Salinization is the process of accumulation of soluble salts such as Sulphates, chlorides, of Ca, Mg, Na & K, in soils in the form of a salty (Salic) horizon. ❖ The intensity and depth of accumulation vary with the amount of water available for leaching. ❖ As a result of the accumulation of salts, Solonchak or Saline soils. ❖ It is quite common in Arid & Semi-arid regions. ❖ It may also takes place through capillary rise of saline ground water and by inundation of sea water in marine and coastal soils. ❖ Salt accumulation may also results from irrigation and seepage in areas of impeded drainage (low lying areas).	❖ Desalinization is the process of removal, by leaching, of excess soluble salts from horizons or soil profile (that contained enough soluble salts to impair the plant growth). ❖ It is effected by leaching of soluble salts from soil, either with rain water or with irrigation water of good quality. ❖ Drainage (natural / artificial) is highly essential for desalinization. ❖ So long as there is excess of soluble salts, the soils remain aggregated and allow water to pass through, because the Na saturated colloids do not hydrolyze in the presence of excess calcium during leaching.

6. **Alkalization Vs Solonization Vs Solodization / Dealkalization**

Alkalization Vs Solonization	Solodization / Dealkalization
❖ It is the process of accumulation of sodium ions on the exchange complex of the clay, resulting in the formation of Sodic / Alkali soils. ❖ The equivalent Russian terminology is Solonetz. ❖ As the soil dries, most of the Ca and Mg are precipitated from solution as carbonates much before Na is precipitated. ❖ As a result, the concentration of Na in the soil solution increases which react with exchange sites and adsorbed on exchange sites. ❖ The net result is the increase in pH to 9.0 or above. ❖ Being Na as deflocculating agent, it disperses the soil particles and leads to poor physical condition (structure – Prismatic / Columnar) of soil. ❖ The organic matter is dissolved under alkaline conditions and thus forms black organo-clay coating on the ped-faces, and therefore such Sodic soils are also termed as Black Alkali soil. ❖ Common in Indo-gangetic plain of Punjab, Haryana and Uttar Pradesh.	❖ It is the process of removal of Na from the exchange sites of the clay. ❖ Addition of Gypsum as amendment will results in replacement of Na by Ca from the exchange sites. ❖ Nax + CaSO4 $\longrightarrow$ Na_2SO_4 + Cax, whrere, X - clay exchange sites. ❖ As the Na_2SO_4 formed is easily soluble, they are leached out by providing drainage. ❖ The added calcium (flocculating agent) helps in improving the physical condition (structure) of the soil. ❖ The resulting soils are known as Soloth and they are very much like the Zonal soils with which these are associates and have a light colored surface horizon over a heavy dark brown coloured B-horizon.

<u>Write a short note on the following</u>

1. **Humification**
    - Humification is the one of the important fundamental/ basic Pedogenic processes.
    - It is the process of decomposition of organic matter and synthesis of new organic substances.
    - It is the process of transformation / decomposition of raw organic matter into humus (complex) with the help of microorganisms.
    - Helps in formation of surface humus layer (Ao).
    - Percolating water passing through Ao dissolves certain organic acids and affects the development of A and B horizons.
    - Under forests in a temperate climatic zone, there is accumulation of undecomposed as well as humified organic matter.
2. **Gleization (Ag, Bg, Cg)**
    - The term 'Glei' is of Russian origin which means Blue, Grey, Green clay.
    - It is a process of refers to reduced condition (anaerobic) that leads to give Blue, green / grey clay.
    - The reduction of Fe compounds to soluble ferrous forms is responsible for the production of typical bluish / greyish horizon.
    - It is a process of soil formation resulting in development of a Glei (or Gley horizon) in the lower part of the profile above the parent material due to poor drainage condition (lack of oxygen) and where waterlogged conditions prevail.
    - The process is not particularly dependent on climate but often on drainage conditions.
    - The poor drainage conditions results from lower topographic position, impervious parent material and lack of aeration.
3. **Horizonation**
    - Horizonation is the process of differentiation of soil into different horizons along the depth of the soil body.
    - Horizonation includes the processes by which soil materials are differentiated into several horizons in a soil profile.
    - The soil horizons may be pronounced to be observed visually or may have to be differentiated by means of certain soil characteristics.
    - It is mainly due to the basic processes like Humification, Eluviation, and Illuviation.
4. **Pedodurbation**
    - In contrast to Horizonation, Pedodurbation is the process of mixing of soil body and thereby tend to have a more uniform soil profile vertically (chemical and mineralogical composition is essentially uniform with depth)
    - Intermixing of soil horizons takes place in certain soil due to external factors.
    - May be Faunal (Animals), Floral (Plants), Argilli (churning / swelling and shrinkage clay).
5. **Argillation**
    - Argillation is the process that results in leaching of dispersed clay particles from upper to lower horizons, giving rise to a textural horizon.
6. **Lessivage**
    - Dachaufour, 1977, a French scientist first used the term 'Lessivage' to describe the mechanical movement of clay and iron oxides in suspension from the A horizon without undergoing chemical alteration.
    - Consequently a clay depleted eluvial horizon (E, formerly termed as A2) develops.
    - The process implies a certain level of leaching of divalent cations as the cause of deflocculation for mobilizing clay.

VII. Component of Soil / Composition of Soil

Fill in the blanks

1. Soil may be described as a three-phase system, composed of ________________, _____________ and ______________ phases.
2. Soils having more than _____________of organic constituents are designated as Organic soils.
3. If the inorganic constituents dominate in the soils, they are called as ___________ soils.
4. The majority of the soils of India are ______________ soils.
5. The four major components of soil are _____________, _________________, _____________ and __________________.
6. The solid, liquid and gaseous phase of soil is represented by ________________________, ____________________ and ______________ respectively.
7. Soil Air and Soil water bear ________________________ relationship with each other.
8. The relative proportion of the mineral matter, organic matter, soil air and soil water is _______, ____________, _________ and __________ respectively.
9. The water containing dissolved organic and inorganic solutes is called as __________________.
10. The ____________________ constituents form the bulk of the solid phase of soil.
11. The organic matter content of most of the Indian soils is in the range of __________________.
12. The air-filled pores constitute the _______________ phase of the soil system.
13. The water-filled pores constitute the _______________ phase of the soil system.
14. The CO_2 concentration of soil air is ____________ than the atmospheric air, whereas the oxygen concentration of soil air is slightly _____________ than the atmospheric air.
15. The ________ phase provides mechanical support for and nutrients to plants of the soil system.
16. The __________ phase supplies water and along with it dissolved nutrients to plant toots of the soil system.
17. The aeration need of plants is satisfied by _________________ phase of the soil system.
18. The inorganic constituents of soil consist of mainly ______________ & _______________.
19. At any stage, the sum of the volumes of the liquid and gaseous phases remains __________ for a particular soil.
20. Soil contains about ______________ solid space and _________________ pore space.

Describe the following

1. Explain briefly the significance / role of major components of soil.

Answers – Components of Soil / Composition of Soil

Fill in the blanks

1. Solid, Liquid and Gaseous
2. 20 %
3. Mineral
4. Mineral
5. Mineral matter, Organic matter, Soil Air, Soil Water
6. Mineral & Organic matter, Soil Water and Soil Air
7. see-saw / inverse
8. 45 % , 5 %, 25 % and 25 %
9. Soil solution
10. Inorgainc
11. 1 - 2 %
12. Gaseous
13. Liquid
14. much higher, lesser
15. Solid
16. Liquid
17. Gaseous
18. Primary & Secondary minerals
19. constant
20. 50 % and 50 %

VII. Soil Physical Properties – Soil Texture

Fill in the blanks

1. The relative propotai0on of various sized soil particles of a given soil is ________________.
2. The size of Clay particles (diameter) is less than _____________, while the size of the silt particle is ranged from _______ to ______.
3. The sedimentation velocity of the falling particles (spherical) with same density is directly proportional to ______________ and inversely proportional to _________________.
4. _______________ is considered as an important intrinsic permanent (non-dynamic) physical property of soil.
5. _________________ and _________________ are the important methods of determination of soil texture (mechanical analysis).
6. _____________, _____________ and _____________ are the soil separates of soil.
7. A soil particle having size of less than 2 micron is known as __________________.
8. If the size of particle is equal or less than 1 micron, then it is known as ________________.
9. A soil having high proportion of sand shows less ________________ capacity and poor in soil ____________________ status.
10. Clay particles are ___________ charged, whereas the cations are ______________ charged.
11. One micron being equal to __________ mm, the highest limit is __________ for coarse sand.
12. ____________ is considered as the most reactive fraction of the soil which is mainly due to large _____________ area.
13. The surface area of the soil particles are ____________ proportional to the size of the particles.
14. Separation of soil particles with the help of sieves or sieve analysis is called as _____________.
15. The sedimentation technique of soil textural analysis follows ___________ law.
16. Particle size distribution analysis is otherwise called as _________________.
17. The textural classes of the soil are identified with the help of ______________________.
18. Heavy soils are so called because of the _______________________________________.
19. The heavy soils are ____________ in weight per unit volume and having_____________ porosity than the sandy soils.
20. When the soil is rubbed between thumb and fingers, preferably in the wet condition, sand feels ______________, silt feels ___________________ when dry and clay feels _________________ when ________________ when dry.
21. ISSS refer to _________________________, while USDA refers to ________________________.
22. The relatively inert ______________ and _____________ fractions can be called as the soil 'skeleton', while the _______ can be regarded as the 'flesh' of the soil and all the three fractions of the solid phase constitute the ___________ of the soil.
23. A _____________ soil is often considered to be optimal soil for plant growth and contains a balanced mix of sand, silt and clay.
24. In USDA textural triangle, out of the ________ textural classes, the clayey soil must contain at least ____________ of clay separates.
25. In sandy soils, the sand and clay separate comprises ___________ & ____________of the material by weight.

Differentiate the following

1. Sandy soil Vs Clayey soil
2. Assumptions of Stoke's Law Vs Limitations of Stoke's Law
3. Heavy soils Vs Light Soils
4. Loamy Sand Vs Sandy Loam

Write a short note on the following

1. Stoke's Law
2. Soil Texture
3. Mechanical Analysis / Particle size distribution analysis
4. Classification of soil separates (ISSS and USDA)
5. Textural Triangle and Textural Classes

Describe the following

1. What is Stoke's Law and give a brief account of its assumptions and limitations.

Answers – Soil Physical Properties – Soil Texture

Fill in the blanks

1. Soil texture
2. 0.002 mm (2 µm / 2microns), 0.002 mm (2 microns) to 0.02 mm (20 microns)
3. the square of the radius of the particle (r^2), viscosity of the medium (η)
4. Soil texture
5. International Pipette method and Bouyoucos Hydrometer method
6. Sand, Silt, and Clay
7. Clay
8. Soil (clay) colloids
9. Water holding, Aggregation
10. Negatively, Positively
11. 10^{-3} mm, 2 mm
12. Clay, specific surface
13. inversely
14. True Particle Analysis
15. Stoke's Law
16. Mechanical Analysis
17. Textural Triangle (USDA / ISSS)
18. greater power that is needed to plough them than to plough sandy soils
19. lighter, higher
20. coarse and gritty, floury / powdery, very sticky & plastic, very hard
21. International Society of Soil Science (renamed as International Union of Soil Sciences), United States Department of Agriculture
22. Sand and Silt, Clay, Matrix
23. Loam (Loamy)
24. 12, 35 %
25. > 70 % & < 15 %

Differentiate the following

1. **Sandy Soil VS Clayey Soils**

Sandy Soil	Clayey Soils
❖ It is one of the broad groups of soil textural class of coarse in nature. ❖ It is again subdivided into different subgroups based on the relative proportion of sand, silt and clay. ❖ In sandy soils, the sand separate comprises more than 70 % of the material by weight and less than 15 % of the material is clay. ❖ There are two specific textural classes are recognized viz. Sand, Loamy Sand. ❖ In general, Sandy soils have low water holding capacity, high bulk density, well aerated, low organic matter content, little or no swelling and shrinkage, poor aggregation status, good drainage condition and high leaching of nutrients and pollutants. ❖ Sandy soils are coarse-textured soils and termed as light soils because of the lesser power that is needed to plough the soil, but heavier than clay soils in terms of weight per unit volume.	❖ It is one of the broad groups of soil textural class of fine in nature. ❖ It is again subdivided into different subgroups based on the relative proportion of sand, silt and clay. ❖ IN clayey soils, the characteristics of clay separate are distinctly dominant. ❖ Three distinct specific textural classes exists, Sandy Clay (having at least 35 % Clay), Silty Clay (at least 40 % Clay) and Clay (at least 40 % Clay). ❖ The clayey soils have high water and nutrient holding capacity, low bulk density, poorly aerated, very slow drainage unless cracked, high to medium in organic matter content, medium to excellent nutrient supplying capacity, good aggregation capacity and high swelling and shrinkage. ❖ Clayey soils are fine-textured soils and termed as heavy soils because of the greater power that is needed to plough the soil, but lighter than sandy soils in terms of weight per unit volume.

2. **Assumptions Vs Limitations of Stoke's Law**

Stoke's Law - Assumptions	Stoke's Law - Limitations
❖ Particles must be spherical.	❖ The effect of different particle shapes (irregular, flake, round, sub-rounded, rod, plate and disc, etc.) on the settling velocities of clay particles is a major limitation for the accuracy of this law. Soil colloidal particles are mostly plate shaped and fall slower than spherical particles of the same mass.
❖ The particles must be smooth and rigid.	❖ It is highly probable that the soil particles are not completely smooth over their entire surfaces and not rigid. To make the law valid, an equivalent spherical radius is so defined that particles having this radius fall with the same velocity as the soil particles.
❖ It is necessary to maintain a known constant temperature, so that convection currents are not set up. And the rate of fall of particles varies inversely with the viscosity of the medium which changes with change in temperature.	❖ The reliability of the result also depends on the maintenance of a known constant temperature during the mechanical analysis based on the principle.
❖ The particles must be of uniform density.	❖ The density of all the soil particles is not same and depends on the mineralogical and chemical composition of the particles and their degree of hydration. To make the law valid, the average value of 2.65 g / cm^3 for particle density is accepted.
❖ Fall of particles must be unhindered and not affected by then proximity (very near) of the wall of the vessel or of the adjacent particles. The extent of liquid must be great (sufficiently diluted) in comparison with the size of the particles.	❖ Particles falling very near the wall of the vessel (0.1 mm distance) are slowed down in their descent. Many falling particles may drag finer particles along with them.
❖ Size of the particles must be large compared to the size of the molecules of the liquid, so that the medium can be considered homogeneous, i.e. no Brownian movement occurs and will not affect the fall or the particles must be big enough to avoid Brownian movement.	❖ The larger limit of particles exhibiting Brownian movement is approximately 0.0002 mm. The colloidal particles less than 0.0002 mm are found to exhibit Brownian movement and so that the rate of falling varied.
❖ There must be no slipping between the particles and liquid.	❖ Many fast falling particles may drag finer particles along with them.
❖ The velocity of the falling particles must not exceed a certain critical value so that the viscosity of the medium remains the only resistance to the fall of the particles.	❖ Particles greater than silt size fraction of a soil mass can not be separated accurately with the help of this law.
❖ The viscosity of the medium must be constant.	❖ The viscosity of the medium may vary as the viscosity of the medium is if there is any variation in temperature during mechanical analysis.
❖ The suspension must be still without any turbulence or existence of laminar / viscous flow of particles. Any turbulence will alter the velocity of fall of particles.	❖ Particles > 0.08 mm diameter settle / sediment quickly and cause turbulence therefore, are analyzed by other procedure.

3. **Heavy Soils Vs Light Soils**

Heavy Soils	Light Soils
❖ In field terminology, clayey soils are often termed heavy soils. ❖ Heavy soils are so called because of the greater power that is needed to plough them than to plough sandy soils. ❖ In contrast, the heavy soils are lighter (high porosity) than sandy soils in terms of weight per unit volume. ❖ Heavy soil contains higher proportion of clay separate in comparison to silt and sand separates.	❖ In field terminology, sandy soils are often termed light soils. ❖ Light soils are so called because the energy required for tillage is much less in light soils than heavy soils. ❖ In contrast to the term 'light', the light soils are dense (low porosity) and heavier than sandy soils in terms of weight per unit volume. ❖ Light soil contains higher proportion of sand separate in comparison to silt and clay separates.

4. **Loamy Sand Vs Sandy Loam**

Loamy Sand	Sandy Loam
❖ Loamy sand is one of the specific textural classes under the broad group of Sandy soils. ❖ In loamy sand, sand separate comprises more than 70 % of the material by weight and less than 15 % of clay.	❖ Loamy sand is one of the specific textural classes under the broad group of Loamy soils. ❖ Basically sandy loam is a loamy soil with the dominance of sand separate. ❖ To qualify for the modifier sandy, a loamy soil must have at least 40 to 45 % of sand separate.

<u>Write a short note on the following</u>

1. **Stoke's Law**
   - ❖ G.G.Stokes (1850) suggested the relationship between the radius of a particle and its rate of fall in a liquid. He stated that the resistance offered by the liquid to the fall of the particle varied with the square of the radius of the spherical particles and not with the surface.
   - ❖ According to Stoke's Law, the viscous force given by $6\Pi r\eta v$ which balances the gravitational force given by $4/3\ \Pi r^3(d_p\text{-}d_l)g$ acting on a single particle of equivalent spherical radius, r.
   - ❖ Thus, $6\Pi r\eta v = 4/3\ \Pi r^3(d_p-d_l)g$ and therefore $v = 2/9\ gr^2(d_p-d_l) / \eta$,
     Where, v is sedimentation velocity (cm sec^{-1})
     g is acceleration due to gravity (cm sec^{-2})
     r is equivalent spherical radius of the particle (cm)
     d_p is density of the particles (g cm^3)
     d_l is density of the liquid / fluid (g cm^3)
     η is absolute viscosity of the suspending fluid (g sec cm^{-1} or poise)
   - ❖ In case of systems where d_p, d_l, η and g are constant at specified temperature, the above equation simplifies to $v = K\ r^2$, where K is constant.
   - ❖ Thus, the sedimentation velocity being directly proportional to the square to the radius of the particle and inversely proportional to the viscosity of the suspending medium.
   - ❖ This law is the basis for the determination of soil texture by sedimentation technique method.
2. **Soil Texture**
   - ❖ Soil texture is one of the important physical properties (mechanical behavior) of a soil.
   - ❖ Soil texture refers to the relative proportion / percentage by weight of various soil separates viz. Sand, Silt and Clay.
   - ❖ It refers to the relative proportion of particles of various sizes viz. sand, silt and clay in a given soil.
   - ❖ As the soil texture can not be altered / changed, it is considered as permanent and intrinsic basic physical property of soil.
   - ❖ Soil texture can be determined by mechanical analysis and soil textural class can be identified further with the help of Textural triangle diagram (USDA / ISSS).

3. **Mechanical Analysis / Particle size distribution Analysis**

- ❖ Mechanical analysis is the analytical procedure by which the particles are separated into various size groups from the coarsest sand, through silt, to the finest clay.
- ❖ The determination of relative distribution of the ultimate or individual soil particles below 2 mm in diameter is called as Mechanical Analysis.
- ❖ It involves two steps viz. separation of all particles from each other (complete dispersion into ultimate particles) by using various dispersing agents like hydrogen peroxide (OM), HCl (lime and sequioxides) and NaOH / sodium hexametaphosphate / Calgon solution (other binding agents) and measuring the amount of individual soil separates (coarser fractions – sieving technique i.e coarse sand – 2 mm sieve, finer sand – 70 mesh sieve, finer fractions – sedimentation technique).
- ❖ There are several methods of mechanical analysis, of which the International Pipette method and the Bouyoucos Hydrometer method are very important.

4. **Classification of soil separates**

- ❖ The size groups are arbitrarily chosen, for instance, by the International Society of Soil Science (ISSS), as clay (less than 2 microns / 0.002 mm), silt (2 – 20 microns / 0.002 mm – 0.02 mm), fine sand (20 –200 microns / 0.02 – 0.2 mm), and coarse sand (200 – 2000 microns / 0.2 – 2 mm). (one micron is equal to 10^{-3} mm).
- ❖ In United States of Department of Agriculture (USDA) system, sand fractions are further subdivided into 5 subgroups.

ISSS System

Clay	Silt	Fine Sand	Coarse Sand

0.002 mm 0.02 mm 0.2 mm 2 mm

USDA System

Clay	Silt	Very Fine Sand	Fine Sand	Medium Sand	Coarse Sand	Very Coarse Sand

0.002 0.05 0.1 0.25 0.5 2 mm

5. **Textural Triangle and Textural Classes**

- ❖ Each side of the equilateral triangle represents percent by weight of clay, silt and sand.
- ❖ Different textural classes are indicated in the textural triangle.
- ❖ The textural class of a given soil with specific sand, silt and clay percentage is identified by following way.
- ❖ For a soil having x % of clay, y % silt and Z % of sand, point X is located on the clay co-ordinate, an imaginary line is drawn parallel to sand co-ordinate; then from then point Y on the silt co-ordinate, an imaginary line parallel to the clay co-ordinate is drawn; then from the point Z on the sand co-ordinate, an imaginary line parallel to the silt co-ordinate is drawn. The textural; class in the diagram in which the point of intersection lies is the textural class of the soil.
- ❖ If the point of intersection lies exactly in the boundary line between textural classes, of which usually a finer textural class is preferred as a textural class of the soil.

VIII. Soil Physical Properties – BD, PD & PS____

Fill in the blanks

1. The Apparent Specific Gravity of soil is otherwise called as ________________________.
2. The Absolute Specific Gravity of soil is otherwise called as ________________________.
3. When Pore Space of soil increases, the BD of soil _________________.
4. The Bulk Density of the soil ranged from _____________ to ________________.
5. The Average Particle Density of soils is _______________ and is due to _________________.
6. The Unit of Bulk & Particle Density is _________________.
7. ____________ & ________________ are the few examples of heavy minerals.
8. The Compaction of soil _____________ Pore Space of soil and _____________ BD of soil.
9. The Micro pores and Macro Pores are otherwise called as _____________________ & __________________ respectively.
10. The Capillary Pores are responsible for ___________________ and the Non-capillary Pores are responsible for _____________________.
11. The percentage of soil volume occupied by Pore Space is termed as _______________.
12. The formula for finding out of Porosity of soil using BD & PD is _______________.
13. An ideal soil should have equal proportion of ________ & ________ pores.
14. The increase in clay content of soil increases the proportion of __________ pores.
15. The addition of organic matter _____________ the porosity and _____________ the BD of soil.

Write a short note on the following

1. Bulk Density / Apparent Specific Gravity
2. Particle Density / Absolute Specific gravity / True Density
3. Pore Space and Porosity
4. Interrelationship between BD, PD and Porosity
5. Factors affecting BD
6. Factors affecting PD
7. Factors affecting PS
8. Capillary Porosity
9. Non-capillary Porosity
10. Total Porosity

Answers – Soil Physical Properties – BD, PD & PS

Fill in the blanks

1. Bulk Density
2. Particle Density / True Density
3. decreases
4. 1.0 g /cc or g /cm^3 or Mg/m^3 to 1.8 g/cc or g /cm^3 or Mg/m^3
5. Soil texture
6. 2.65 g /cc or g /cm^3 or Mg/m^3, dominance of Quartz (2.60), Feldspar (2.65), Mica (2.5 – 2.75) in soil
7. Haematite, Pyrite
8. reduces, increases
9. Capillary & Non-capillary
10. Water retention, Water Movement
11. Porosity
12. [1 – (BD/PD)] x 100
13. micro and macro
14. micro pores
15. increases, decreases

IX. Soil Physical Properties – Soil Colour

Fill in the blanks

1. The Red Colour of the Soil is mainly due to the presence of _______________.
2. The Munsell Colour Variables / Notations are __________, ___________& ___________.
3. The Brown Colur of the soil is mainly due to the presence of the mixture of __________ & _____________.
4. On wetting, the soil Colur becomes _______________.
5. The Munsell Colur Variable ‘Chroma’ 0 & 8 indicates _____________ and _______________ respectively.
6. The Munsell Colur Variable ‘Value’ 0 & 10 indicates _____________ and _______________ respectively.
7. The presence of mottled colour in soil indicates the ________________________________.
8. The Bluish and or Greenish Colour of the soil is an indication of ______________________.
9. ______________________ is used to identify the soil Colour.
10. In 2.5 YR 5/4, ____________, ________ & ____________ are the expressions of Hue, Value and Chroma.

Write a short note on the following

1. Munsell Colour Notations / Variables
2. Hue
3. Value
4. Chroma
5. Factors affecting the Soil Colour
6. Effect of hydration on Soil Colur
7. Composition of Soil and Soil Colour
8. Munsell Colour Chart
9. Pedochromic Vs Lithochromic
10. Significance of Soil Colour

Answers – Soil Physical Properties – Soil Colour

Fill in the blanks

1. Iron Compounds
2. Hue, Value & Chroma
3. Organic Matter & Iron Oxides
4. Darker
5. Neutral Grey, More Brighter Colour
6. Black, White
7. Alternate Wetting and Drying
8. Reduction / Reduced condition of soil
9. Munsell Colour Chart
10. 2.5 YR – Hue, 5 – Value, 4 - Chroma

Write a short note on the following

1. **Munsell Colur Notations / Variables**
   - ❖ Soil colour is the result of light reflectance and described in term of Munsell Color Notations / variables / parameters as follows:
   - ❖ **Hue** – Hue refers to the dominant spectral colour of the soil. It is prefixed with numbers from 0 to 10 at the interval of 2.5. For example, in Hue 2.5 YR, YR indicates the Yellow Red. As the prefixing number increases from 0 to 10, redness decreases and yellowness increases.
   - ❖ **Value** – Value refers to the relative lightness or darkness of the spectral colour and is the function of total amount of reelected light. It is indicated by the numbers ranging from 0 to 10. '0' indicates black colour and 10 indicates white colour. As the number increases from 0 to 10, the colour becomes brighter or changes from dark to light.
   - ❖ **Chroma** – Chroma refers to the relative purity / strength of the spectral colour lightness or darkness of the soil colour and is the function of total amount of reelected light. It is indicated by the numbers ranging from 0 to 8. '0' indicates neutral grey and 8 indicates more vivid / pure colour. As the number increases from 0 to 8, the colour becomes more vivid / pure or the strength of the colour increases.

2. **Hue**
   - ❖ Hue is one of the Munsell colour notations / variables to denote the soil colour in Munsell colour chart.
   - ❖ Hue refers to the dominant spectral colour (like Y, YR, R, GY) of the soil. It is prefixed with numbers from 0 to 10 at the interval of 2.5. For example, in Hue 2.5 YR, YR indicates the Yellow Red. As the prefixing number increases from 0 to 10, redness decreases and yellowness increases.

3. **Value**
   - ❖ Value is one of the Munsell colour notations / variables to denote the relative lightness or darkness of the soil colour in Munsell colour chart.
   - ❖ Value refers to the relative lightness or darkness of the spectral colour and is the function of total amount of reelected light. It is indicated by the numbers ranging from 0 to 10. '0' indicates black colour and 10 indicates white colour. As the number increases from 0 to 10, the colour becomes brighter or changes from dark to light.

3. **Chroma**
   - ❖ Chroma is one of the Munsell colour notations / variables to indicate the relative purity of spectral colour in Munsell colour chart.
   - ❖ Chroma refers to the relative purity / strength of the spectral colour lightness or darkness of the soil colour and is the function of total amount of reelected light. It is indicated by the numbers ranging from 0 to 8. '0' indicates neutral grey and 8 indicates more vivid / pure colour. As the number increases from 0 to 8, the colour becomes more vivid / pure or the strength of the colour increases.

3. **Factors affecting the Soil Colur**
   - ❖ **OM** – High OM imparts black to dark brown colour to the soil.
   - ❖ **Fe Compounds** – Presence of Fe compounds imparts red, brown, yellow tinges to the soil.

- **Si, Li and other salts** – Presence of silica, lithium and other salts give white or light colour to the soil.
- **Mixture of OM & R_2O_3** – The brown colour of the soil is mainly contributed to the presences of the mixture of OM and sesquioxides in soil.
- **Alternate Wetting & Drying** – Alternate wetting and drying leads to mottled variegation in soil due to oxidation and reduction of Fe and Mn compounds.
- **Oxidation and Reduction** – Under reduced conditions, soil colour becomes bluish or greenish due to the reduction of Fe compounds from ferric to ferrous. Under oxidized educed conditions, soil colour becomes more brighter (reddish) due to the dominance of ferric ions in soil.
- **Hydration –** Upon hydration, the soil colour changes from reddish to yellowish gradually via reddish brown and yellowish brown.

X. Soil Physical Properties – Soil Structure

Fill in the blanks

1. The arrangement of sand, silt and clay particles within the soil is termed as _______________.
2. The distinction and durability of the Ped is described in terms of _______________.
3. The 'Clod' refers to _________________ and whereas the 'Ped' refers to __________________.
4. If the faces and edges of the Block-like soil structure are round, it is termed as ___________.
5. The Granular structure is having relatively __________ porous than Crumby Structure.
6. The Columnar structure is frequently observed in _____________ soils.
7. The addition of organic matter ______________ the soil structure.
8. The presence of Na-clay in soil ______________ the soil structure, whereas the presence of Ca-clay _________________ the soil structure.
9. If the top of the Ped of Prism-like structure is round and smooth, it is termed as ____________________.
10. An artificially formed soil mass is termed as ______________________.

Write a short note on the following

1. Soil Structure – Definition
2. Types of Soil Structure
3. Classes of Soil Structure
4. Grades of Soil Structure
5. Factors influencing Soil Structure
6. Classification of Soil Structure
7. Importance of Soil Structure in plant growth
8. Flocculation / Aggregation
9. Deflocculation / Dispersion
10. Genesis of Soil Structure

Describe the following

1. Define Soil Structure. How Soil Structure is related with the plant growth?
2. Give a brief account about the classification of soil structure.

Answers – Soil Physical Properties – Soil Structure

Fill in the blanks

1. Soil Structure
2. Grade
3. An artificially formed soil mass, Natural Aggregates
4. Sub Angular Blocky
5. non-porous / less porous
6. Sodic / Alkali
7. improves
8. destroys, improves
9. Columnar
10. Clod

XI. Soil Air

Fill in the blanks

1. The soil volume that is not occupied by the soil particle is termed as _________________.
2. ______________ & _______________ are the mechanisms involved in the gaseous exchange between Soil Air and Atmospheric Air.
3. When the Oxygen level in Soil Air becomes less than _________, the crop growth is highly restricted.
4. _______________ is the principal mechanism of gaseous exchange that accounts about 90 %.
5. The Mass flow is mainly due to the _____________________, whereas the Diffusion is mainly due to the ________________ of individual gases.
6. The Diffusion of gases between Soil Air and Atmospheric Air follows ___________ law.
7. The Soil Air has ______ oxygen than Atmospheric Air, due to ______________________.
8. The Soil Air has ______ carbon dioxide than Atmospheric Air, due to ______________________.
9. The sub-surface soil has relatively ____________ oxygen than the surface soil.
10. On cropping, the carbon dioxide concentration in soil Air ________________.
11. Soil Air contains higher level of ___________________, but lower level of ___________ gas.
12. The Micro Pores are otherwise called as _____________________ and are responsible for ______________.
13. The Macro Pores are mainly responsible for ________________ and otherwise termed as ________________.
14. The volume of gaseous phase varies ____________ with the liquid phase of the soil.
15. The Soil Aeration refers to ____.

Write a short note on the following

1. Soil Aeration
2. Intercrumb Pores Vs Intracrumb Pores
3. Air Capacity
4. Mass Flow
5. Diffusion
6. Factors Influencing Soil Air
7. Composition of Soil Air Vs Atmospheric Air
8. Fick's Law
9. Renewal of Gases Between Soil Air and Atmospheric Air
10. Significance of Soil Aeration

Answers – Soil Air

Fill in the blanks

1. Pore Space
2. Mass Flow and Diffusion
3. < 10 %
4. Diffusion
5. total pressure gradient, partial pressure gradient
6. Fick's Law
7. less, depletion of oxygen in Soil Air by plant roots and microorganisms
8. more, release of carbon dioxide by plants and microorganisms through respiration
9. less
10. increases
11. CO_2, O_2
12. Capillary Pores / Intracrumb Pores, Water Retention
13. Non-capillary Pores / Intercrumb Pores, Water Movement
14. inversely
15. the constant movement of air in the soil mass resulting in the renewal of gases between soil air and atmospheric air.

XII. Soil Temperature

Fill in the blanks

1. Solar Radiation penetrates the soil profile largely by ______________________.
2. _________________ is the most predominant source of Soil Heat.
3. In humid region, the major portion of Global Radiation loses through the process of __________________.
4. In winter, the Precipitation increases the Soil Heat / Temperature, because of the _________ Specific Heat of Precipitation / water.
5. The amount (cal / g) of heat energy consumed during the process of Evaporation (conversion of Liquid from to Gaseous form of water) is __________.
6. The cooling action of Precipitation during Summer is due to the ______ temperature of rain water than the soil.
7. The low temperature of wet soil is partly due to ________________ & __________________.
8. The amount of heat energy required to raise the temperature of 1 g of any substance by 1˚C, is termed as ____________________.
9. The Specific Heat of Water, Dry Soil, Mineral Matter, Quartz, Iron, Organic Matter, & Air is ______, ______, ______, ______, ______, ______ & ______ respectively.
10. The amount of heat energy required for water to raise the temperature is much ____________ than warm soil solids.
11. The unit of Heat Capacity is __________, whereas the unit of Specific Heat / Specific Heat Capacity is __________.
12. Moist Soils are always cooler, because of the ______ specific heat than the Dry Soils.
13. When the Heat Capacity is expressed per unit volume, it is called as ______________________.
14. The Heat Capacity (Cs) is calculated by multiplying the Specific Heat (C) with the________________.
15. The Thermal Conductivity (k) is expressed in terms of _____________________ and found to be _________________ in Sandy soils than Clayey soils.
16. Thermal Diffusivity (Dh) is the ration of Thermal Conductivity (k) to the ___________________ and expressed in terms of _____________________.
17. The Spheroidal structure of soil warm more _________ than the Platy type of soil structure.
18. The Heavy-textured clayey soils absorbs the heat more _________ than the Coarse-textured sandy soils.
19. The Soil Temperature in ridged topography is _____________ than the leveled one.
20. The Compaction __________ the Density of soil and thereby _____________the Thermal Conductivity and Heat Capacity of soil.
21. Spreading of Black-coloured substances like ________________ leads to ______________ in the Soil Temperature, whereas the spreading of white coloured substances like _____________ results in ___________ in the Soil Temperature.
22. The tillage operations lead to _______ the Porosity of soil and _________ the Thermal Conductivity and thereby the Soil Temperature.
23. Addition of Organic Matter to the soil ____________ the Specific Heat and ___________ the Soil Temperature.
24. The product of Specific Heat and Density is termed as _________________________.
25. The removal of excess water from soil (drainage) ___________________ the Specific Heat and _______________ the Soil Temperature.
26. The temperature of soils of Temperate regions is always _______________ than the soils of Tropical region.
27. The Soil Temperature is always __________ than the Atmospheric Temperature.
28. The bare soils warm ____________ than the soils with vegetation.
29. The soils containing much more mineral matter get heated __________ than those soils containing higher amount of Organic Matter.
30. ___________ has the lowest Specific Heat of major soil constituents, and _____________ has the highest, excepting water.

Write a short note on the following

1. Sources of Soil Heat
2. Losses of Soil Heat
3. Specific Heat
4. Heat Capacity and Volumetric Heat Capacity
5. Heat of Vaporization
6. Thermal Conductivity
7. Thermal Diffusivity
8. Factors influencing Soil Temperature
9. Measurement of Soil Temperature
10. Management of Soil Temperature

Describe the following

1. Discuss different factors which influence the soil temperature.
2. What are the sources and losses of soil heat?
3. Discuss the role of soil temperature in soil fertility.

Answers – Soil Temperature

Fill in the blanks

1. conduction
2. solar radiation
3. evaporation
4. high
5. 580
6. low
7. evaporation, high specific heat of water
8. Specific Heat
9. Water - 1.00 cal/g , Dry Soil - 0.2 cal/g, Mineral Matter - 0.181 cal/g, Quartz - 0.18 cal/g, Iron - 0.11 cal/g , Organic Matter - 0.462 cal/g, & Air - 0.24 cal/g
10. higher
11. cal [C (cal/g) x m (g)], cal/g
12. high
13. Volumetric Heat Capacity
14. mass (m)
15. Cal/˚C/cm/sec or J//˚C/cm/sec, higher
16. Volumetric Heat Capacity (Cv), cm^2/sec
17. quickly
18. slowly
19. higher
20. increases, increases
21. Charcoal Powder, increases, Chalk Powder, decreases
22. increases, reduces
23. increases, decreases
24. Volumetric Heat Capacity
25. reduces, increases
26. cooler
27. higher
28. quickly
29. easily
30. Quartz, Humus

XIII. Soil Water

Fill in the blanks

1. At Filed Capacity, the macro pores are filled with ____________, but the micro pores are filled with ______________.
2. A soil having high proportion of sand shows less _____________ capacity and poor in soil ___________ status.
3. 100 K Pa is equivalent to _________ Atmospheric Bars.
4. If the water is held by the soil with a suction of 1000 cm column of water, the equivalent pF value is _________.
5. _______________ developed the concept of pF scale.
6. As the pF value increases from 0 to 7, the soil water content ________________.
7. The equivalent pF value under the condition of soil moisture at Field Capacity, Max. WHC, Saturated Soil, Wilting Point, Hygroscopic Point, Air-dry and Over-dry is ________, ________, _________, _________, ________, _________, & __________ respectively.
8. The equivalent Atmospheric Pressure under the condition of soil moisture potential at Field Capacity, Max. WHC, Saturated Soil, Wilting Point, Hygroscopic Point, Air-dry and Over-dry is ________, ________, _________, ________, _______, _______, & _______ respectively.
9. As the soil water content increases, the soil moisture potential ____________.
10. ___________ is used as a source of Gamma Ray in 'Gamma Ray Attenuation' technique of soil moisture determination.
11. Thermalization refers to ____.
12. The volume wetness can be calculated by multiplying the Mass Wetness with ____________.
13. The formulae for calculating the % moisture content in soil on oven dry basis is ___________.
14. __________ method is the simplest and most widely used method for measuring soil moisture.
15. ________________ is the source of fast neutrons in 'Neutron Scattering Method' of soil moisture estimation.

Write a short note on the following

1. Available Water Vs Hygroscopic Water
2. Capillary Water VS Gravitational Water
3. Field Capacity Vs Wilting Coefficient
4. Percolation Vs Seepage
5. Cohesion Vs Adhesion
6. Surface Tension
7. pF Scale
8. Soil Water Potential Vs Soil Water Tension / Suction
9. Free Energy of Soil Water
10. Max. Water Holding / Retention Capacity
11. Wilting Coefficient Vs Hygroscopic Coefficient
12. Moisture Equivalent Vs Field Capacity
13. Available Water Capacity
14. Permeability Vs Hydraulic Conductivity
15. Percolation Vs Infiltration

Describe the following

1. Briefly explain the movement of soil water.
2. Discuss the various methods of determination of Soil Moisture.
3. Define Soil Moisture Constants. Briefly outline the different Soil Moisture Constants.
4. Write in brief about the role of Soil Water for crop production. Briefly discuss the factors affecting the Soil Water.

Answers – Soil Water

Fill in the blanks

1. Air, Water
2. Water Holding Capacity, Moisture / Aggregation
3. 0.98 / 1
4. 3
5. Schofield (1935)
6. decreases
7. 2.53 - Field Capacity, 1.00 - Max. WHC, 0 - Saturated Soil, 4.18 - Wilting Point, 4.50 - Hygroscopic Point, 6.00 - Air-dry and 7.00 - Over-dry
8. 1/3 or 0.33 - Field Capacity, 0.01 - Max. WHC, 0.001 - Saturated Soil, 15.0 - Wilting Point, 31.0 - Hygroscopic Point, 1000.0 - Air-dry and 10000.0 - Over-dry
9. increases
10. Cs^{137}
11. Conversion of fast neutrons into slow neutrons
12. bulk density of soil
13. [(Ww- Wd) / Wd] x 100, where, Ww – weight of the moist / wet soil, Wd – weight of the dry soil
14. Gravimetric
15. Mixture of α emitter (Radium / Americium) & Beryllium

XIV. Soil Consistency

Fill in the blanks

1. The degree and kind of _____________ & __________________ decides the soil consistency.
2. _______________ is considered as the dynamic physical property of soil.
3. The soil consistency when wet, is described in terms of ______________ & _______________.
4. The ability of the soil to be moulded in to any shape under applied force is termed as ________.
5. As the water content in soil increases from dry to moist, the cohesion _________________.
6. The Upper Plastic Limit is otherwise called as _______________, whereas the Lower Plastic Limit is otherwise termed as _____________________.
7. The index / number / range over which soil is plastic, is called as _________________.
8. The moisture content below and above the UPL, the soil consistence is _____________ & ________________ respectively.
9. The Scouring Point lies just _______________& _______________ the UPL in highly plastic soil and slightly plastic soil respectively.
10. The formula for calculating the COLE / PVC is ____________________.
11. The consistency of soil is ________________ when dry, _____________ when moist and ___________ when wet.
12. The moisture at which the soil water mass no longer sticks into the foreign objects / steel spatula is called as ________________.
13. The Shrinkage Limit refers to ____.
14. The Sticky Limit refers to ____.
15. The Cohesion refers to ____, whereas the Adhesion refers to ____.
16. Soils containing more than about ___________ % clay exhibit plasticity.
17. Attraction of soil solids for water provides _________________ force.
18. The quality of adhesion of soil to other objects is described in terms of ___________________.
19. _______________, the Swedish Agriculturist divided the entire cohesive range from solid to liquid in terms of water content.
20. The value of COLE > 0.03, indicates the presence of significant amount of __________________________ type of clay.

Write a short note on the following

1. Soil Consistency / Soil Consistence
2. Atterberg's Limit
3. COLE / PVC
4. Liquid Limit / UPL Vs Plastic Limit / LPL / Friable Limit
5. PI / PN and Soil Plasticity

Describe the following

1. What is Soil Consistency? Explain in brief the soil consistency at different moisture conditions.

Answers – Soil Consistency

Fill in the blanks

1. Cohesion and Adhesion
2. Soil Consistency
3. Stickiness and Plasticity
4. Plasticity
5. decreases
6. Liquid Limit, Plastic Limit / Friable Limit
7. Plasticity Index / Plasticity Number
8. Plastic , Liquid
9. below, above
10. (Lm / Ld) – 1
11. hard, friable, plastic
12. Scouring Point
13. the lower limit of volume change at which there is no further reduction in volume as the water is evaporated. < Shrinkage limit, the soil is dry and hard; between shrinkage limit and LPL, the soil is moist and friable.
14. the moisture content above which a mixture of soil and water adhere to a steel spatula or other objects, when soil wets with water.
15. the attraction of substances of like characteristics such as that of one water molecule for another, the attraction of substances / materials of unlike characteristics such as the attraction between the soil and water molecule.
16. 15 %
17. Adhesive
18. Stickiness
19. Atterberg
20. Expanding (Like Montmorillonite)

Write a short note on the following

1. **Soil Consistency**
    - ❖ Soil consistency / soil consistence is defines as the physical manifestations of the physical forces of cohesion and adhesion acting within the soil at various moisture constants.
    - ❖ Soil consistency refers to the degree and kind of cohesion and adhesion of soil materials, at varying moisture conditions.
    - ❖ Soil consistency is the behavior of soil under stress.
    - ❖ Soil consistency is the term used by the soil scientists to describe the resistance of a soil to mechanical stress or manipulations at various moisture contents.
    - ❖ Soil consistency can also be defined as the resistance of a soil to rupture / soil's resistance to penetration by an object.
2. **Atterberg's Limits**
    - ❖ Atterberg (1911) divided the entire cohesive range from solid to liquid in term of water content as follows:
        1. Upper Plastic Limit (UPL) / Liquid Limit
        2. Lower Plastic Limit (LPL) / Plastic Limit / Friable Limit
        3. Plasticity Index (PI) / Plasticity Number
        4. Shrinkage Limit
        5. Sticky Point
        6. Scouring Point
        7. Upper Plastic Limit / Liquid Limit
    - ❖ **Upper plastic limit** (pF – 0.5) – It represents the moisture content of soil at a point where the soil water mass just flows under applied force and fails to retains its shape. Moisture content above the UPL, a mixture of soil and water flow as viscous liquid and below which, the soil is plastic.
    - ❖ **Lower plastic limit** (pF 2.8 to 3.3) – It refers to the moisture content of a soil at a point where its consistence changes from plastic to friable. The soil water mass is unable to

change the shape continuously under applied force and ultimately the mass is broken into fragments. The moisture content below the LPL, the soil starts to crumble when rolled into a thread under the palm of hand. It is the upper limit of moisture content fro tillage operations for most of the crops except Rice.

- ❖ **<u>Plasticity Index</u>** – It refers to the range over which a soil remains plastic. It is the difference between the moisture contents of a soil at its UPL and LPL. Different soils are characterized by a specific PI. Soils with high PI are difficult to plough (example Smectitic clay like Montmorillonite clay).). It is otherwise called as Plastic Number and obtained by the formula of PI = MC at UPL – MC at LPL.
- ❖ **<u>Shrinkage Limit</u>** – It is the lower limit of volume change at which there is no further decrease in volume as water is evaporated. Cohesive soils expand in volume when wetted with water and shrink when the water is evaporated or removed. Below the shrinkage limit, the soil is dry and hard; between shrinkage limit and LPL, the soil moist and friable; Between LPL and UPL, the soil is plastic.
- ❖ **<u>Sticky Limit</u>** – It is the moisture content above which a mixture of soil and water will adhere to a steel spatula or other objects which when wets with water.
- ❖ **<u>Scouring Point</u>** – It is the moisture content at which the soil water mass no longer sticks to a foreign object. It lies below the UPL for the highly plastic soils and lies above the UPL for the slightly plastic soils.

3. **COLE / PVC**

- ❖ COLE refer to the Coefficient Of Linear Extensibility.
- ❖ It is otherwise called as Potential Volume Change (PVC).
- ❖ Certain soils like Black soils (Montmorillonite clay) have the capacity to swell significantly when wet and to shrink and crack when dry.
- ❖ This property is quantified by determining the COLE / PVC and calculated by using the formula of COLE = (Lm – Ld) – 1, where, Lm – Length of wet soil, Ld – Length of dry soil.
- ❖ If COLE is > 0.09 in a soil, significant swelling and shrinkage is expected in that soil.
- ❖ If COLE is > 0.03, it indicates that the presence of significant amount of Montmorillonite type of clay in soil.

4. **Upper Plastic Limit (UPL) / Liquid Limit Vs Plastic Limit / LPL / Friable Limit**

- ❖ **<u>Upper plastic limit</u>** (pF – 0.5) – It represents the moisture content of soil at a point where the soil water mass just flows under applied force and fails to retains its shape. Moisture content above the UPL, a mixture of soil and water flow as viscous liquid and below which, the soil is plastic.
- ❖ **<u>Lower plastic limit</u>** (pF 2.8 to 3.3) – It refers to the moisture content of a soil at a point where its consistence changes from plastic to friable. The soil water mass is unable to change the shape continuously under applied force and ultimately the mass is broken into fragments. The moisture content below the LPL, the soil starts to crumble when rolled into a thread under the palm of hand. It is the upper limit of moisture content fro tillage operations for most of the crops except Rice.

5. **Plasticity Index (PI) Vs Soil Plasticity**

- ❖ **<u>Plasticity Index</u>** – It refers to the range over which a soil remains plastic. It is the difference between the moisture contents of a soil at its UPL and LPL. Different soils are characterized by a specific PI. Soils with high PI are difficult to plough (example Smectitic clay like Montmorillonite clay). It is otherwise called as Plastic Number and obtained by the formula of PI = MC at UPL – MC at LPL.
- ❖ **<u>Soil Plasticity</u>** – It refers to the capacity of a soil to be moulded into any shape and the property of toughness. When stress is applied, the soil changes its shape and keeps the changed shape when the stress is removed from soil. Soils with clay content < 15 % do not exhibit plasticity in any moisture range.

XV. Soil Colloids

Fill in the blanks

1. 1 : 1 type of Clay Minerals are composed of one _____________ and one ___________ sheet.
2. Dispersion of clay is owing to the __________ of negatively charged particles for each other.
3. Other factors being equal, higher the CEC of a soil, _____________ is the buffering capacity.
4. If the size of the particle is equal to or less than 1 μ, then it is known as __________ particle.
5. Clay particles are __________ charged, whereas the cations are ___________ charged.
6. ___________ is an example of Amorphous Clay Minerals.
7. __________, ___________, _____________ & _____________ is an example of 1 : 1, 2 : 1 Expanding, 2 : 1 Non-expanding, 2 : 2 / 2 : 1 : 1 type of clay minerals.
8. Dimorphic Clay refers to _______________________, whereas the Trimorphic Clay refers to _____________________.
9. The magnitude of the charge is termed as ______________ and the permanent charge of the clay is mainly due to ____________________________.
10. The Brownian Movement of the soil colloids is mainly attributed to ______________________.
11. The shape of the clay colloids in soil is _______________________.
12. The Crystalloids are ____________ in size when compared to the colloidal particles.
13. The development of pH-dependant / variable charges is as a result of _____________________, which are most dominant in ____________ colloids and __________ with increase in pH.
14. The Zero Point of Charge (pHo) is otherwise called as _____________ and refers to ______________________________.
15. In Isomorphous substitution of ions, Si^{4+} is replaced by __________ in STH, and Al^{3+} is replaced by __________ in AOH layer.

Write a short note on the following

1. Soil Colloids
2. Brownian Movement
3. Surface Area and Surface Charge
4. Isomorphous Substitution / Isomorphous Substitution of ions
5. pH-dependant charges VS pH-independent charges of clay
6. Organic Colloids
7. Layer Silicate Clays (LSC)
8. R_2O_3 Clays Vs Amorphous Clays
9. Ionic Double Layer
10. STH Vs AOH Layer

Describe the following

1. What is Soil Colloids? Explain in brief about the colloidal properties of soil colloids / soil clay.
2. How charges are being developed on the surface of the clay? or What are the sources of development of positive and negative charges on clay surface?

Answers – Soil Colloids

Fill in the blanks

1. STH, AOH
2. repulsion
3. higher
4. colloidal / clay colloidal
5. negatively, positively
6. Allophane
7. Kaolinite, Montmorillonite, Illite, Chlorite
8. the layer silicate clay consists of 2 layers viz. one STH and one AOH, the layer silicate clay consists of 3 layers viz. two STH and one AOH
9. Zeta Potential, Isomorphous substitution
10. the collision between the soil clay particles and water molecules in which they are dispersed
11. plate / flake / lamella like
12. smaller
13. the dissociation of exposed OH groups in the crystal edges / ionisable hydrogen ions from the exposed OH groups in the crystal edges of Si & Al, organic colloids, increases
14. isoelectric point, the pH at which the net surface charge is zero or the pH at which the positive charges equals the negative charges
15. Al^{3+}, Mg^{2+}

XVI. Ion Exchange

Fill in the blanks

1. The Acid Soils are otherwise called as ____________________________.
2. The % BS is ___________________ in Acid Soils.
3. The formula for calculating the % BS and % Base Unsaturation is ______________ & _____________ respectively.
4. As the pH increases, the % BS _________________________.
5. One meq / 100 g of soil is equivalent to ______ c.mol (p^+) / kg of soil.
6. As the clay content of the soil & pH increases, the CEC of soil _______________.
7. The AEC _____________ with increase in soil pH and ________________ with increase in soil acidity.
8. The role of Anion Exchange is of most significant in relation to __________________________.
9. The order of power of replacement of anions on positively charged clay surface is ___________________________________.
10. P Fixation / Reversion of Phosphate refers to _____________________________________.

Write a short note on the following

1. Cation Exchange Capacity / CEC
2. % Base Saturation / % BS
3. Anion Exchange Capacity / AEC
4. P Fixation
5. Power of replacement of cations on clay surface
6. Cation Exchange Vs Anion Exchange
7. Ion Exchange
8. Significance of Cation Exchange
9. Significance of Anion Exchange
10. CEC / AEC ratio

Answers – Ion Exchange

Fill in the blanks

1. Base Unsaturated Soils
2. low
3. % BS = (Bases or Basic Cations / CEC) x 100, % BUS = (Acidic Cations / CEC) x 100 or % BUS = 100 - % BS
4. increases
5. one
6. increases
7. decreases, increases
8. Phosphate ions and their fixation
9. $OH^- > H_2PO_4^- > SO_4^{2-} = Cl^- > NO_3^-$
10. the conversion of the soluble (available) phosphate ions into insoluble (unavailable) hydroxy phosphates of Fe / Al / Ca

Write a short note on the following

1. **Cation Exchange Capacity**
   - Cation Exchange capacity of a soil is defined as the capacity of the colloidal complex (negatively charged) to exchange all its cations with the cations of the electrolyte solution / soil solution.
   - It also represents the total cation adsorbing capacity of a soil.
   - Cation Exchange capacity is expressed in terms of meq / 100 g of soil or c.mol (p^+) / kg of soil.
   - Each clay mineral / soil is characterized by its capacity to exchange the cations called CEC.
   - Cation Exchange capacity in most soil increases with pH.
   - However, at very low pH value, the CEC is higher and at high pH, the CEC is relatively lower.
   - CEC (T) = S + H, where, S = the sum of the basic cations, H = the sum of acidic cations, T = total cations.
2. **% Base Saturation / % BS**
   - The proportion of the cation exchange capacity occupied by the bases / base forming cations is called the percentage base saturation.
   - The cations like Ca, Mg, Na, K are the base forming cations in the soil.
   - A definite correlation exists between the percentage base saturation of a soil and its pH.
   - As the base saturation is reduced as a result of loss calcium in drainage, the pH is also lowered in a definite proportion.
   - As CEC (T) = S + H, % BS = (S / T) x 100 and % BUS = (H / T) x 100 or [(T – S) / T] x 100.
3. **Anion Exchange Capacity / AEC**
   - Anion Exchange capacity of a soil is defined as the capacity of the colloidal complex (positively charged) to exchange all its anions with the anions of the electrolyte solution / soil solution.
   - It also represents the total anion adsorbing capacity of a soil.
   - Anion Exchange capacity is expressed in terms of meq / 100 g of soil or c.mol (e^-) / kg of soil.
   - Each clay mineral / soil is characterized by its capacity to exchange the anions called AEC.
   - Anion Exchange capacity in most soil decreases with pH or increases with the increase in soil acidity. The lower the pH, the greater is the anion adsorption.
   - In general the relative order of anion exchange is $OH^- > H_2PO_4^- > SO_4^{2-} = Cl^- > NO_3^-$
4. **P Fixation**
   - The phenomenon of anion exchange assumes importance in relation to the phosphate ions and their fixation.
   - The adsorption of phosphate ions by clay particles from soil solution reduces its availability to plants. This is known as P fixation.

- ❖ As the reaction is reversible, the phosphate ions again become available when they are replaced by OH ions released by the substances like lime applied to correct soil acidity. Hence, the fixation is temporary.
- ❖ The whole phosphate adsorbed by clay is, however, not exchangeable, as even at pH 7.0 and above.
- ❖ The phosphate ions not only interact with silicate clays, but also with R_2O_3 clays and thereby converting the soluble (available) phosphate ions into insoluble (unavailable) Fe / Al hydroxyl phosphates.
- ❖ If the reaction takes place at a low pH under strongly acid conditions, the phosphate ions are irreversibly fixed and are totally unavailable for the use of plants.

5. **Power of replacement of cations on clay surface**
   - ❖ The power of replacement of cations varies with the type of ion, size, degree of hydration, valency, concentration, kind of clay.
   - ❖ The order of power of replacement of monovalent ions is $H^+ > Cs^+ > Rb^+ > K^+ > Na^+ > Li^+$.
   - ❖ The order of power of replacement of divalent ions is $Ba^{2+} > Sr^{2+} > Ca^{2+} > Mg^{2+}$.
   - ❖ The order of power of replacement of ions in mixed ionic soil system is $H^+ = Al^{3+} > Ca^{2+} > Mg^{2+} > K^+ >$ Na. It means that the Na^+ is more easily replaced than K^+, K^+ is more easily replaced than Mg^{2+} and H^+ behaves like a trivalent ion and therefore more tenaciously bonded by soil colloids.

5. **Cation Exchange Vs Anion Exchange**
   - ❖ The phenomenon of exchange of cations between the soil colloids (negatively charged) and the soil solution is known as cation exchange or base exchange.
   - ❖ The process of migration of cations between the two soil phases in contact (i.e. solid and solution phase) is termed as cation exchange.
   - ❖ The cations that take part in this reaction are called as exchangeable cations (Ca^{2+}, Mg^{2+}, Na^+, K^+, H^+, Al^{3+}).
   - ❖ The phenomenon of exchange of anions between the soil colloids (positively charged) and the soil solution is known as cation exchange or base exchange.
   - ❖ The anions that take part in this reaction are called as exchangeable anions (OH^-, $H_2PO_4^-$, SO_4^{2-}, Cl^-, NO_3^-)
   - ❖ Both cation exchange and anion exchange processes are reversible and the exchange takes place on chemically equivalent basis.

5. **Ion Exchange**
   - ❖ Ion exchange is defined as a reversible process by which the cations and anions are exchanged between the soil colloids (solid phase) and the soil solution (liquid phase) and between the solid phases in close contact with each other, on chemically equivalent basis.
   - ❖ The phenomenon of exchange of cations between the soil colloids (negatively charged) and the soil solution is known as cation exchange or base exchange.
   - ❖ The cations that take part in this reaction are called as exchangeable cations (Ca^{2+}, Mg^{2+}, Na^+, K^+, H^+, Al^{3+}).
   - ❖ The phenomenon of exchange of anions between the soil colloids (positively charged) and the soil solution is known as cation exchange or base exchange.
   - ❖ The anions that take part in this reaction are called as exchangeable anions (OH^-, $H_2PO_4^-$, SO_4^{2-}, Cl^-, NO_3^-)
   - ❖ Both cation exchange and anion exchange processes / reactions are reversible and the exchange takes place on chemically equivalent basis.

5. **Significance of Cation Exchange**
   - ❖ Cation exchange reactions are of greater importance in agriculture.
   - ❖ It influences the retention and liberation of plant nutrients to plants.
   - ❖ It prevents the cations from leaching especially in high rainfall regions.
   - ❖ It controls the soil structure and crumb formation.
   - ❖ It is also responsible for imparting the stable soil structure.
   - ❖ It controls the process of soil formation through the weathering of minerals and formation of clay minerals.
   - ❖ It plays a major role in reclamation of problematic soils like acid and alkali / sodic soils.
   - ❖ It also influences the effect of fertilizers and fertilizer practices.

5. **Significance of Anion Exchange**

- The process of anion exchange assumes importance in relation to phosphate ions and their fixation.
- The anion exchange process controls the availability of phosphate ions to the plants.
- It prevents the anions from leaching to some extent especially in high rainfall regions.
- All anions are not adsorbed equally and readily.
- Some anions such as $H_2PO_4^-$ are adsorbed very readily at all pH values in the acid as well as alkaline range.
- Cl^- and SO_4^{2-} ions are adsorbed slightly at low pH but none at neutral soil, while the NO_3^- ions are not adsorbed at all.
- Hence, at the pH commonly prevailing in cultivated soils, the NO_3^-, Cl^- and SO_4^{2-} are easily lost by leaching. CEC / AEC ratio

5. **CEC / AEC ratio**

- The ration of CEC to AEC is an important index in identification of clay minerals.
- Each clay mineral is characterized by a specific ratio.
- For example, the ration is 6.7 for Montmorillonite, 2.3 for Illite and 0.5 for Kaolinite.
- The ratio gives an idea about the cation and anion exchange capacity of the dominant clay mineral present in the soil.

XVII. Soil Organic Matter

Fill in the blanks

1. Decomposition of organic matter is a _______________ process.
2. The average moisture content and dry matter of plant residues is ________ & ________ respectively.
3. The carbon and oxygen occupies about _________ & __________ percentage in the dry matter of plant residues.
4. The most compound that is highly resistant to microbial attack in the plant residues is ________.
5. _____________ & ______________ tissues are the sources of soil organic matter.
6. Ammonification refers to ____.
7. Nitrification refers to ____.
8. Denitrification refers to ____.
9. Aminization refers to ____.
10. Mineralization refers to ____, while the Immobilization refers to ____.
11. Nitrobactera refers to __________________ & _________________________.
12. The enzyme, Maltase is otherwise called as ____________________.
13. The enzyme, Cellobiase is otherwise termed as __________________________.
14. The end products of carbonaceous compounds of soil organic matter under aerobic conditions are ________________ & ____________________.
15. Starch is hydrolyzed to ____________________ by the action of amylases.
16. _____________ & ______________ will increase the denitrification loss of nitrogen.
17. The decomposition of cellulose in neutral and alkaline soils proceeds more ___________ than acid soils.
18. Hemicellulose decomposes __________________ than cellulose.
19. The order of rate of decomposition of Cellulose, Hemicellulose, and Starch during soil organic matter decomposition is ___________________________.
20. _________________, ________________ & ______________ are the humic substances of humus.
21. Humus contains on an average ____________ percentage of carbon, whereas the carbon content of plant tissues / residues is ____________.
22. Three percent Hydrogen Peroxide has been found to be oxidise the ____________, leaving the undecomposed organic matter un attacked.
23. The C:N ratio of Humus and soil organic matter is ____________ & ____________ respectively.
24. The cation exchange capacity of humus / organic matter far ___________ than that of most silicate clays.
25. The humic compounds and non-humic compounds make up about _________ & __________ percentage of the organic matter respectively.
26. With increase in oxygen content in soil, the loss of nitrogen by denitrification ______________.
27. The form of nitrogen that is specially lost via denitrification is ________________.
28. The end product of denitrification is _________________.
29. The process of conversion of organic N to inorganic N is called as ___________________.
30. The colloidal humus particles are composed of primarily _______, _______ & ______ elements.
31. The water holding capacity of humus on mass basis is ________ times higher than LSC.
32. ____________ is the organism which converts the NO_3 to NO_2 in soil.
33. _________ is the organic compound in SOM that is having the lowest/ slowest rate of decomposition.
34. ____________________ is the microorganism involved in the mineralization of sulphur / oxidation of sulphur.
35. Most of the nitrogen in the soils is present in the ___________ from.
36. The nitrogen content of Indian soils (surface) ranging from ______ to ______.
37. Decomposition of wheat straw in the soil causes immobilization of organic N, because the C : N ratio of wheat straw is _________________.
38. The first step of nitrification is mediated by ________________.
39. ________________ emission in most common in water-logged / rice soils.
40. The process of nitrification that takes place in soil is a ____________ oxidation process.

Write a short note on the following

1. Soil Organic Matter Vs Humus
2. Humic Substances
3. Nitrification Vs Denitrification
4. Aminization Vs Ammonification
5. Mineralization Vs Immobilization
6. Mineralization of Organic S
7. Mineralization of Organic P
8. Proteolysis
9. Composition of Plant Residues
10. C : N ratio

Describe the following

1. Discuss the role of soil organic matter on soil fertility.
2. Explain in brief about the soil organic matter decomposition.

Answers – Soil Organic Matter

Fill in the blanks

1. oxidation
2. 75 % & 25 %
3. 44 % & 40 %
4. Lignin
5. plant & animal
6. the conversion / transformation of the organic nitrogenous compounds into ammonia
7. the process of conversion of ammonia to nitrite by Nitrosomonas and then to nitrate by Nitrobacter
8. the process of conversion of Nitrate into Nitrous oxide or gaseous nitrogen
9. the conversion of Proteins into Amino Acids
10. the conversion of complex organic compounds into simple inorganic compounds, the conversion of simple inorganic compounds into complex organic compounds
11. Nitrosomonas & Nitrobacter
12. α glucosidase
13. β glucosidase
14. CO_2 & H_2O
15. Maltose
16. Water-logging / Anaerobic condition & high pH
17. rapidly
18. faster
19. Starch > Hemicellulose > Cellulose
20. Humin, Fulvic Acid & Humic Acid
21. 58 %, 45 – 50 %
22. humus
23. 10 : 1 – 12 : 1, 40 : 1 – 12 : 1
24. exceeds / high
25. 60 – 80 %, 20 – 30 %
26. decreases
27. Nitrate
28. N_2
29. mineralization
30. C, O and H
31. 4 – 5
32. Bacillus / Pseudomonas / Paracoccus
33. Lignin
34. *Thiobacillus thiooxidans*
35. organic
36. 0.03 to 0.06 %
37. too wide
38. Nitrosomonas
39. Methane
40. biological

XVIII. Soil pH and Nutrient Availability

Fill in the blanks

1. Nitrogen availability is considerably reduced when the pH of the soil is below __________ and above __________.
2. The availability of the P is at its maximum when the soil pH is between ______ & ______.
3. The conversion of soluble (available) P into insoluble (unavailable) P is called as __________.
4. The solubility of K is ___________ when the alkalinity of the soil is due to $CaCO_3$.
5. As the pH decreases, the solubility of Fe, Al and Mn ________________.
6. The availability of Mo is at its maximum when the soil pH is above _________.
7. The availability of Mo is reduced under __________ condition of the soil.
8. The availability of the most of the Micronutrients is at their maximum under _________ condition of the soil.
9. In general, the availability of various phosphate ions to plants is considered to follow the order of ________________________________.
10. The availability of Boron _____________ below PH 5.0 and above 7.0, but above pH 8.5, it again ________________.
11. The Ammonifiers and Nitrifiers are active at pH _____________.
12. At low pH, the availability of K, Ca, and Mg _____________________.
13. The pH range of ______ to ________ is the optimum for availability of most of the nutrients.
14. Liming of Acid soil _____________ the P availability.
15. Alternate drying and heating the soil increases the availability of soil _____________.
16. With increase in soil pH, the Molybdenum availability _________________.
17. Fixation of P can be kept at minimum by maintaining the soil pH in the range of ___________.
18. The availability of B, Cu and Zn progressively ____________ as the soil pH increases.
19. ____________ is the most important factor which governs the availability of nutrients in soil.
20. Application of organic matter _________ the availability of Mn by converting the Mn from the manganic state to _________ state.

Answers – Soil pH and Nutrient Availability

Fill in the blanks

1. 5.5, 9.0
2. 6.5 & 7.5
3. P fixation
4. decreased
5. increases
6. 6.5
7. acidic
8. slightly acidic
9. $H_2PO_4^- > HPO_4^{2-} > PO_4^{3-}$
10. decreases, increases
11. 5.5 to 6.0
12. decreases
13. 6.5 to 7.5
14. increases
15. Manganese
16. increases
17. 6 – 7
18. decreases
19. soil pH
20. increases, manganous

XIX. Soil Buffering Capacity

Fill in the blanks

1. The resistance to change in soil pH is called as _________________.
2. The buffering capacity of clayey soils is __________ than that of sandy soils.
3. With increase in organic matter in soil, the buffering capacity of soil ______________.
4. The greater the CEC, the __________ will be the buffering capacity.
5. The amount of liming maternal required for reclaiming the clayey acid soil is ______________ than that of a sandy acid soil.
6. The amount of acid / alkaline material required to change the pH of a clayey soil with high organic matter is ________________.
7. Buffering action refers to ________________________________.
8. A buffer solution is the one which contains the ____________ acidity and __________ alkalinity.
9. ______________ prevents the sudden changes and fluctuations in soil pH and thereby regulating the availability of nutrients in the soil.
10. As the buffering capacity of the acid soil is more, then the LR ____________.

Answers – Soil Buffering Capacity

Fill in the blanks

1. buffering
2. higher / more
3. increases
4. higher / greater
5. higher
6. higher / greater
7. the power to resist a change in pH
8. reserve acidity and reserve alkalinity
9. Buffering
10. increases

XX. Soil Orders, Soil Survey, LCC & Soils of India

Fill in the blanks

1. The number of soil orders in soil taxonomy is ____________.
2. Lowest category of soil taxonomy is _______________.
3. Highest category of soil taxonomy is ___________________.
4. __________ is the soil order that represents the highly weathered red soils.
5. The soil order that is related to the soils of volcanic origin is _______________.
6. _______________ is the soil order which represents the organic / peaty / muck soils.
7. Most of the Black cotton soils of India is coming under the soil order of ____________.
8. Swelling and Shrinkage clay is generally noticed in the soil order of ________________.
9. Gilgai micro relief is one of characteristics of ___________________.
10. Typical desert soils have been classified in the order of______________ and others in ____________.
11. The soils having mollic epipedons like Tarai soils have been put in the order of ____________.
12. The Gangetic Alluvial soils of India have been classified in the soil order of ______________.
13. Most of the Alluvial soils of India have been classified in the orders of ___________, ___________ and _____________.
14. Most of the Saline / Alkali soil of India have been classified in the soil order of _____________.
15. Most of the Lateritic soils have been classified in the order of _____________ and a few under ________________.
16. Lateritic soils are characterized by a _________ silica-sequioxide ratio.
17. Most of the Red soils of India have been classified under _____________.
18. The most dominant group of soils of India covering the largest area is _________________.
19. The most dominant soil order of India covering the largest area is _________________.
20. The order of dominant soils orders of India in terms of area is ___________________.
21. The Tarai soils of U.P, West Bangal and J & K have been classified in the order of _________.
22. The ______________ soils occurring the Indo-Gangetic plains and the Brahmaputra valley of India cover a large area.
23. In detailed soil survey, aerial photographs with a scale of _________ are generally used as base material for preparation of soil maps.
24. In detailed soil survey, the soils are examined at intervals of _________________ or closer, depending upon the soil heterogeneity.
25. __________ pedons are normally examined for every two hectares in detailed soil survey.
26. In reconnaissance soil survey, the examination of pedons is at intervals of ______________, depending upon the soil heterogeneity.
27. Land capability classes from _______ to _______ are suitable fro cultivation, whereas the classes from ________ to ________ are not suitable cultivation.
28. The total number of classes in Land Capability Classification is ___________.
29. The land capability sub-classes are grouped according to the _____________in each sub-class.
30. The land capability class __________ is suitable fro wildlife and watershed.

Write a short note on the following

1. Detailed soil survey Vs Reconnaissance soil survey
2. Soil orders
3. Higher categories Vs Lower categories
4. Vertisols Vs Lateritic soils / Latosols
5. Land capability subclass Vs Land capability units

Describe the following

1. Discuss in detail the land capability classification.

Answers – Soil Orders, Soil Survey, LCC & Soils of India

Fill in the blanks

1. 12
2. Soil Series
3. Order
4. Oxisols
5. Andisols / Andosols
6. Histosols
7. Vertisols
8. Vertisols
9. Vertisols
10. Aridisols, Entisols
11. Mollisols
12. Inceptisols
13. Entisols, Inceptisols, Alfisols
14. Aridisols
15. Ultisols, Oxisols
16. low
17. Alfisols
18. Alluvial soils
19. Inceptisols (29.1 %)
20. Inceptisols (29.1 %) > Entisols (24.3 %) > Alfisols (24.25 %)
21. Mollisols
22. Alluvial
23. 1 : 15000
24. 1/4 to 1/2 km
25. Two to three
26. 3 – 6 km
27. I to IV (suitable for cultivation), V to VIII (unsuitable for cultivation)
28. 8
29. kind of limitation
30. VIII